

AURIX

Microcontroller Infineon Software Architecture

User's Manual

PORT Driver

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2011-03-02	0.1	Initial Version

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1 Introduction

This User Manual provides information regarding the functionality, configuration parameters and API implementation for the Port Driver.

The Port Driver is built according to the AUTOSAR specification, refer [\[2\]](#).

This User Manual is intended to help the user to get acquainted with the Port Driver implementation for the AURIX Family of microcontrollers. It is the documentation that describes how to use the Port Driver.

1.1 Scope

This document addresses the following features of the Port Driver implementation,

- Aurix microcontroller HW peripheral used to realize the Driver
- File structure of the Port Driver
- Configuration Parameters of the Driver
- Parameters published by Port Driver
- API Specification
- Code Snippets illustrating the common use cases.
- Limitations and Assumptions

This documentation applies to the Port Driver Variant: Variant PB.

1.2 Abbreviations

Table 1 Abbreviations

Abbreviation	Explanation
μC	Micro Controller
ALT	Alternate Output Function
API	Application Programming Interface
AUTOSAR	AUTomotive Open System Architecture
DET	Development Error Tracer
DIO	Digital Input/Output
GPIO	General Purpose Input Output
HW	Hardware
LVTTL	Low Voltage Transistor Transistor Logic
I/O	Input/Output
ISR	Interrupt Service Routine
MCAL	MicroController Abstraction Layer
MCU	MicroController Unit
SW	Software
IFX	Infineon Technologies
ID	Identifier
LVDS	Low-Voltage Differential Signaling
CMOS	Complementary Metal–Oxide–Semiconductor
TTL	Transistor Transistor Logic
LS	Least Significant
MS	Most Significant

MSC	Micro Second Channel
LVDSH	High Performance LVDS
LVDSM	Medium performance LVDS
Px_IOCRRn	Port x Input/Output Control Registers
Px_OMR	Port x Output Modification Register
Px_PDISC	Port x Pin Function Decision Control Register
Px_PDR	Port x Pad Driver Mode 0 Register
Px_PCSR	Port x Pin Controller Select Register
Px_LPCR	Port x LVDS Pad Control Register
HE	High End
EP	Efficient Platform

1.3 References

This section lists down all relevant documents for this User Manual.

Refer to the latest version for the documents without the version information.

- [1] Requirements of the PORT driver: AUTOSAR_SRS_PORT_Driver.pdf , Release 4.0, Rev 3
- [2] Specifications of the PORT driver: AUTOSAR_SWS_PORT_Driver.pdf, Release 4.0, Rev 3
- [3] Target Data Sheet : tc29x_um_V1.0.pdf, tc23x_tc22x_um_v1.0.pdf, tc27xC_um_v1.5.pdf, tc26xB_um_v1.3.pdf, tc27xD_um_v2.2.pdf , tc27x_um_V1.3.1.pdf

2 Port Driver Overview

The Port Driver initializes all pins, which are configurable via the Port Registers of the microcontroller. These pins are combined in hardware itself to form groups called ports.

2.1 Software Hardware Mapping

This section introduces the user to the HW features used for implementation. The [Figure 1](#) illustrates the HW peripherals and their interaction with the Driver.

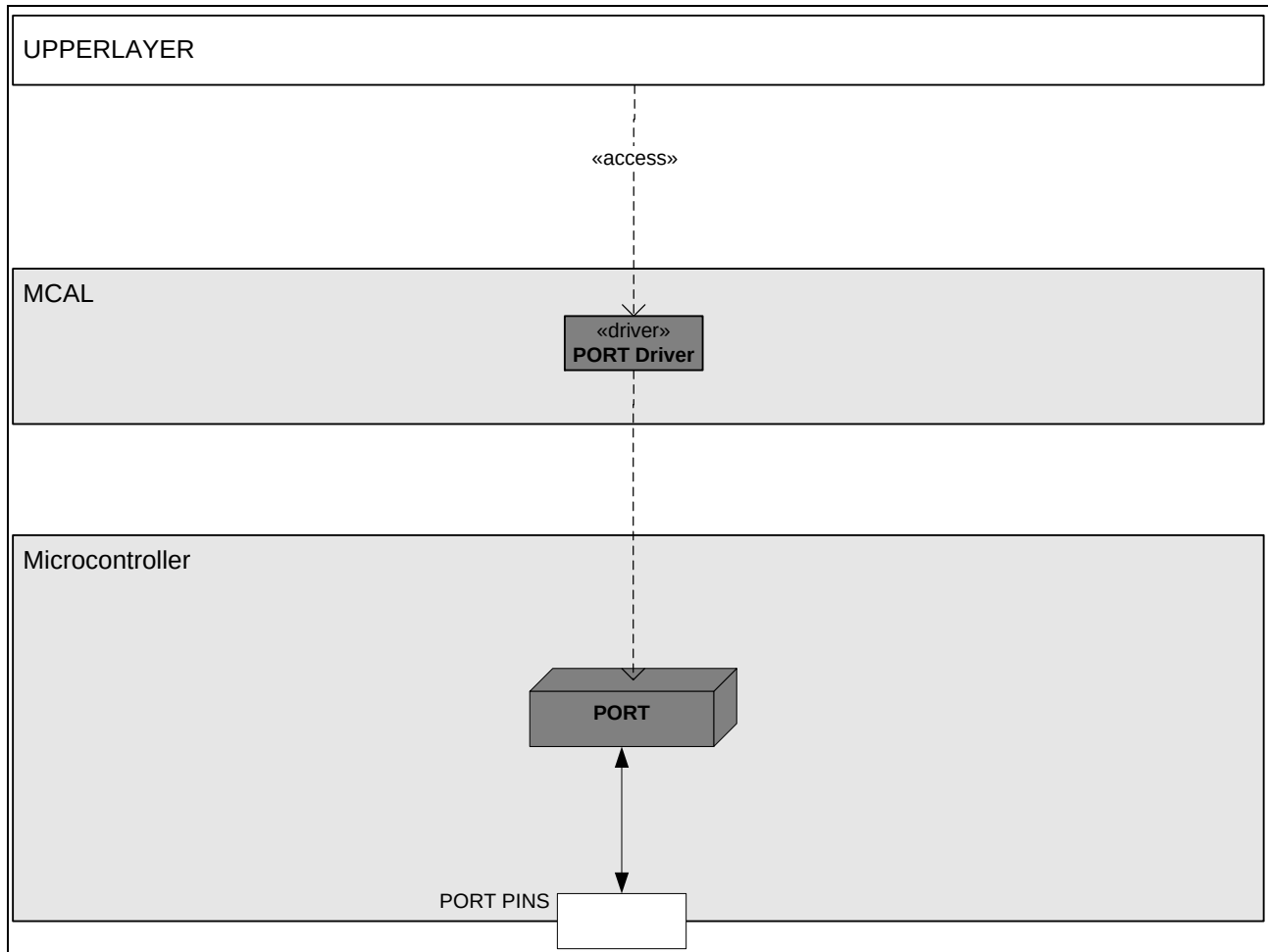


Figure 1 Software Hardware mapping of Port Driver

2.1.1 PORT

The following list summarizes all common features of the Aurix Ports:

- Port pin can be configured as an Input or Output Pin.
- Output driver strength and Slew rate for a Group of pins can be configured.
- The GPIO port lines consists of two classes of pads
 - CMOS pads
 - LVDSH and LVDSM pads [Not applicable for EP devices]
- For LVDSM pads, [Not applicable for EP devices]
 - Voltage supply for the LVDSM PinPair 5V or 3.3V.
- For LVDSH Rx pads, [Not applicable for EP devices]
 - Termination Mode (Internal or External).
 - Signalling Mode (IEEE general Purpose/Special Reduced LVDS link/IEEE Reduced LVDS link)
- For LVDSH Tx Pads, [Not applicable for EP devices]
 - LVDS Power Down or Power up
 - Enable or Disable LVDS pull down resistor.
- The GPIO port lines pad level can be configured
 - CMOS or TTL level (3.3V Pad)
 - Automotive or TTL level (5V Pad)
- Following Input characteristics can be configured
 - Pull-up
 - Pull-down
 - No pull device

- Following Output characteristics can be configured
 - Push Pull
 - Open Drain
- Following Port pin functionality can be configured
 - General purpose Input/Output
 - Output function ALT1
 - Output function ALT2
 - Output function ALT3
 - Output function ALT4
 - Output function ALT5
 - Output function ALT6
 - Output function ALT7
- The port pins used for ADC as analog input pins can also be used as digital input pins.

Many pins are multiplexed for different functionality. If the pin is configured as input pin the functionality of pin is fixed by hardware. If the pin is configured as output pin, user can configure the functionality of pin which is also called alternate function of the pin (ALT1 / ALT2 / ALT3 / ALT4 / ALT5 / ALT6 / ALT7). These alternate functions are referred to as different modes of the pin. Refer the hardware user manual [\[3\]](#) for more details on alternate functions of each pin.

2.2 Software Driver Description

The [Figure 2](#) shows the Application Programming Interface (API) for the Port Driver and its interaction with other software Drivers.

- Development Error Tracer (DET) is required if DET is enabled for the Port Driver.
- Safety reporting will be enabled if safety feature is enabled

APIs of the Driver are explained in detail in chapter [7](#)

The Configuration Data of the Driver is also depicted in the figure. The individual containers and their configuration parameters are explained in detail in chapter [4](#)

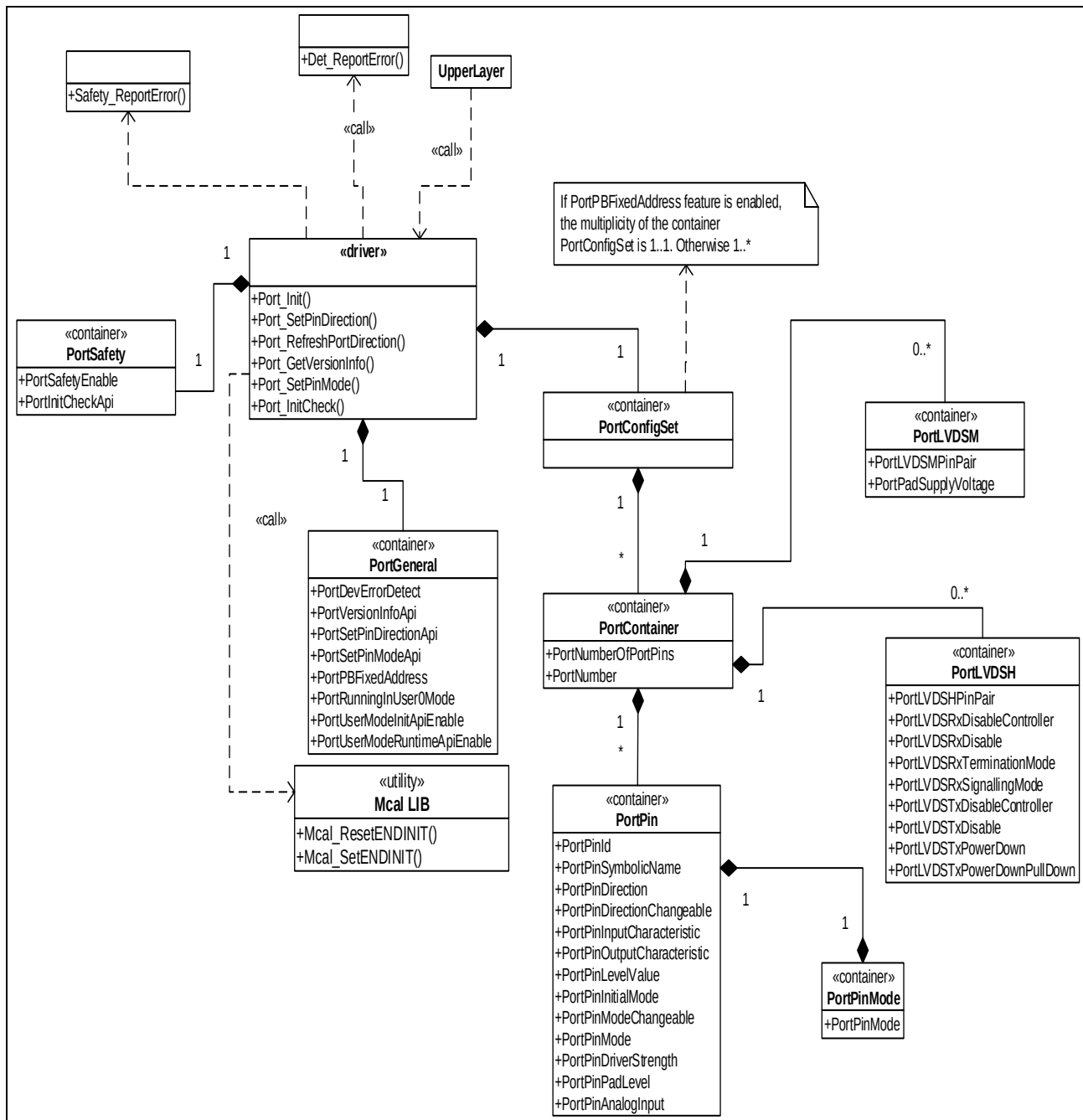


Figure 2 Interfaces of Port Driver

3 File Structure

This section provides details about the Port Driver files and other related files.

3.1 Driver File Structure

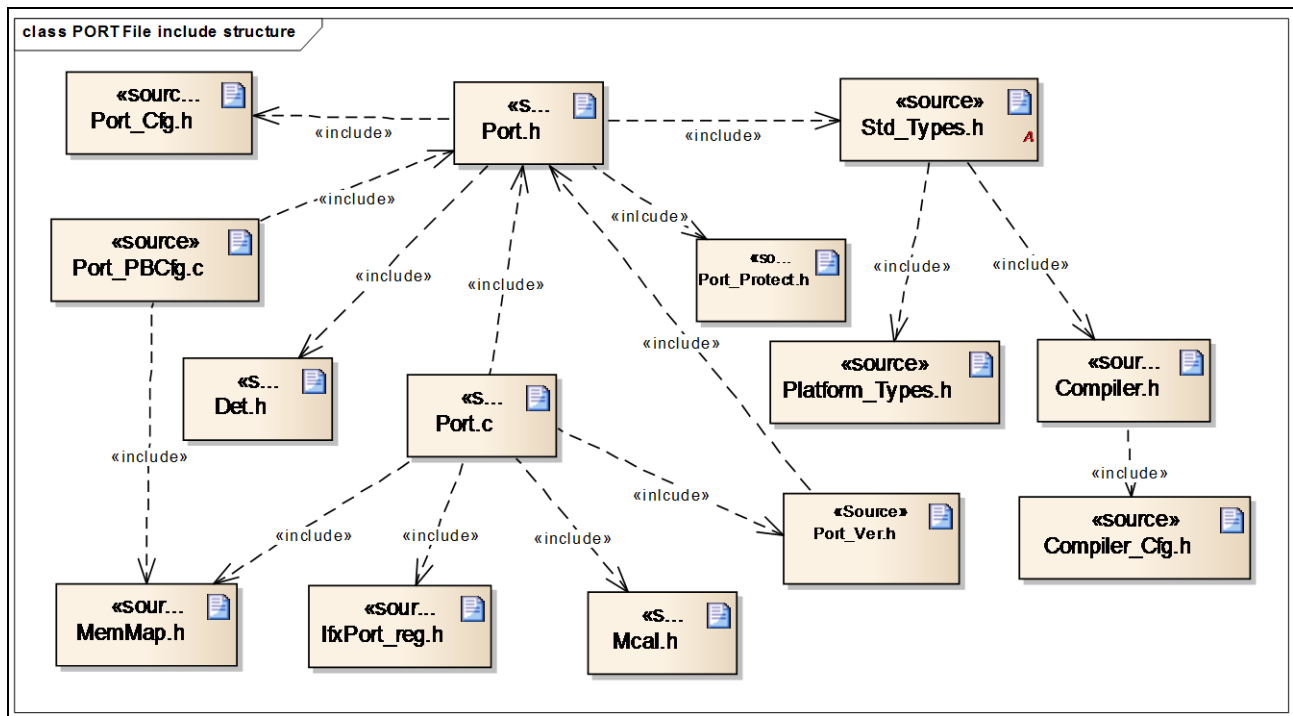


Figure 3 Driver File Structure

Table 2 File Structure

File Name	Description
Platform_Types.h	Platform Dependencies are declared here.
Compiler.h	Compiler Dependencies are declared here.
Compiler_Cfg.h	Contains abstraction of compiler specific keywords used for addressing data and code.
Std_Types.h	Standard Data types to be used are declared here.
IfxPort_reg.h	Includes the PORTS register definition file for microcontroller
Mcal.h	Provides library functions for MCAL layer.
Det.h	Provides the functionality of Development Error Tracer.
MemMap.h	Mapping of code and data (variables, constant variables) to specific memory sections.
Port.h	Exports Port Driver functionality.
Port_Ver.h	Contains AUTOSAR specific (AS403 and AS321) Major and Minor version check and PORT driver SW Major and Minor version check.
Port.c	Implementation of Port Driver functionality. Single file is maintained across autosar specifications AS403 and AS321.
Port_Cfg.h	Contains Driver precompile configuration parameters.
Port_PBCfg.c	Contains Driver postbuild configuration parameters.
Port_Protect.h	Included to define all the first level indirection macros for Protected mode support for supervisory access via OS calls.

4 Getting Started

This section gives you a way of quickly programming the PORT driver for typical functionality.

4.1 Set the direction of a port pin

Configurations are mentioned in Tables 3 and 4; while the sample code is given in Code Listing 1.

The configuration tables describe the modifications required; other values can be left at defaults.

Table 3 PortGeneral Configuration

Configuration Parameter	Configuration Value
PortSetPinDirectionApi	TRUE (PORT_SET_PIN_DIRECTION_API = STD_ON)

Table 4 PortPin Configuration

Configuration Parameter	Configuration Value
PortPinId	0x000
PortPinSymbolicName	PORT_0_PIN_0 [cover parentID=DS_AS4XX_PORT_PR2849] PortPin Symbolic name value according to the Requirement ASW:4397 is deviated. [cover]
PortPinDirection	PORT_PIN_OUT
PortPinDirectionChangeable	TRUE
PortPinInputCharacteristic	PORT_PIN_IN_PULL_UP
PortPinOutputCharacteristic	PORT_PIN_OUT_PUSH_PULL
PortPinLevelValue	PORT_PIN_LOW

Code Listing 1 Programming Sequence for using Port Driver to set the direction of a port pin

```
001:  /* Initialization of PORT Module */
002:  Port_Init(&Port_ConfigRoot[0]);
003:
004:  /* change the direction of a port pin to input */
005:  Port_SetPinDirection(PORT_0_PIN_0, PORT_PIN_IN);
```

4.2 Change the mode of a port pin

Configurations are mentioned in Table 5 and Table 6; while the sample code is given in Code Listing 2.

The configuration tables describe the modifications required; other values can be left at defaults.

Table 5 PortGeneral Configuration

Configuration Parameter	Configuration Value
PortSetPinModeApi	TRUE (PORT_SET_PIN_MODE_API = STD_ON)

Table 6 PortPin Configuration

Configuration Parameter	Configuration Value
PortPinId	0x001
PortPinSymbolicName	PORT_0_PIN_1
PortPinDirection	PORT_PIN_OUT
PortPinOutputCharacteristic	PORT_PIN_OUT_PUSH_PULL
PortPinInitialMode	PORT_PIN_MODE_ALT3
PortPinLevelValue	PORT_PIN_LOW
PortPinModeChangeable	TRUE
PortPinMode	PORT_PIN_MODE_GPIO_ALL.

Code Listing 2 Programming Sequence for using Port Driver to change the mode of a port pin

```

001:  /* Initialization of PORT Module */
002:  Port_Init(&Port_ConfigRoot[0]);
003:
004:  /* Change the mode to ALT5 */
005:  Port_SetPinMode(PORT_0_PIN_1, PORT_PIN_MODE_ALT5);

```

4.3 Get the version info of port driver

Configurations are mentioned in Tables 7; while the sample code is given in Code Listing 3.

The configuration tables describe the modifications required; other values can be left at defaults.

Table 7 PortGeneral Configuration

Configuration Parameter	Configuration Value
PortVersionInfoApi	PORT_VERSION_INFO_API = STD_ON

Code Listing 3 Programming Sequence for using Port Driver to get the version info of port driver

```

001:  /* variable declaration*/
002:  /* Pointer to store Version Info of Port module */
003:  Std_VersionInfoType VersionInfo;
004:
005:  /* Initialization of PORT Module */
006:  Port_Init(&Port_ConfigRoot[0]);
007:
008:  /* Get the Version Information */
009:  Port_GetVersionInfo(&VersionInfo);

```

5 Configuration Documentation

[cover parentID=SAS_NAS_PORT_PR445]

The following chapters summarizes

- All configuration parameters and their description
- Configuration parameter variance w.r.t supported HW.
- The type definition used to represent them in the code.

[cover]

5.1 Configuration Concept

5.1.1 Configuration Class

The development of Basic Software Drivers involves the following development cycles:

- Compiling,
- Linking and
- Downloading of the executable to ECU memory.

Configuration of parameters can be done in any of these process-steps: pre-compile time, link time or even post-build time. According to the process-step that does the configuration of parameters, the configuration classes are categorized as below.

Pre-compile time: The pre-compile time configuration is implemented by means of compiler switches or variables affecting the total build process. Pre-compiler configuration requires the Driver to be compiled, linked and located and hence the values of pre-compile parameters must be **frozen before compilation** and therefore can be used in source code distributions of the Driver only.

Link time: The Link time configuration doesn't affect compilation stage but impacts the linkage and locating stages. The Link time parameters can be used in binary (library) distributions of the Driver and the values of the Link time parameters are specified by means of external configuration source code that must be linked to the Driver during the build process.

Post-build time: The Post-build time configuration will not affect the build process and hence Post-build time configuration is applied to Driver at runtime. Since configuration location is statically independent of the Driver functionality itself but linked dynamically (e.g. at Driver initialization), post-build time configuration is used for re-programmable configuration data.

5.1.2 Configuration Variant

Based on the most suitable configuration class that may be needed for the configuration parameters of the Driver, a Driver can be delivered as the following variants

Variant PC: This variant is limited to pre-compile-configuration parameters only. The intention of this variant is to optimize the parameters configuration for a source code delivery.

Variant LT: This variant allows a mix of pre-compile time-, link time configuration parameters. The intention of this variant is to optimize the parameters configuration for an object code delivery.

Variant PB: This variant allows a mix of pre-compile time-, post build-time configuration parameters. The intention of this variant is to optimize the parameters configuration for a re-loadable binary.

The Port Driver is delivered as Variant PB

5.1.3 How to read the Configuration Class field

The examples cited below help the user to identify the configuration class of the parameter for a given delivery type.

Note: Topics [5.1.1](#), [5.1.2](#) should be read for understanding the configuration class field

Table 8 Configuration Class Example 1

Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--

'x' indicates that the parameter is implemented as a pre-compile parameter in all types of deliveries i.e. Variant PB, Variant LT, Variant PC.

Table 9 Configuration Class Example 2

Configuration Class:	Pre-compile	x	Variant PC
	Link Time	--	--
	Post build	x	Variant PB

'x' indicates that the parameter is implemented as a

1. pre-compile parameter if the delivery type is Variant PC delivery
2. post-build parameter if the delivery type is a Variant PB delivery

5.2 Container: PortGeneral

This container lists down all the general configuration parameters for Port Driver

5.2.1 PortDevErrorDetect

Table 10 PortDevErrorDetect

Name:	PortDevErrorDetect		
Source:	This parameter is defined by the AUTOSAR		
File:	Port_Cfg.h		
Range:	TRUE - PortDevErrorDetect is Enabled FALSE - PortDevErrorDetect is Disabled		
Default Value:	FALSE		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Switches the Development Error Detection and Notification STD_ON or STD_OFF. This configuration parameter is named as PORT_DEV_ERROR_DETECT in the configuration file.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5.2.2 PortVersionInfoApi

Table 11 PortVersionInfoApi

Name:	PortVersionInfoApi		
Source:	This parameter is defined by the AUTOSAR.		
File:	Port_Cfg.h		
Range:	TRUE - Version info API enabled FALSE - Version Info API disabled		
Default Value:	FALSE		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch STD_ON / STD_OFF used to enable / disable the API to read out the modules version information, i.e.,		

	Port_GetVersionInfo() This configuration parameter is named as PORT_VERSION_INFO_API in the configuration file
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.2.3 PortSetPinDirectionApi

Table 12 PortSetPinDirectionApi

Name:	PortSetPinDirectionApi		
Source:	This parameter is defined by the AUTOSAR.		
File:	Port_Cfg.h		
Range:	TRUE - Port_SetPinDirection() API is enabled FALSE - Port_SetPinDirection() API is disabled		
Default Value:	FALSE		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch STD_ON / STD_OFF used to enable / disable the use of the API Port_SetPinDirection(). This configuration parameter is named as PORT_SET_PIN_DIRECTION_API in the configuration file		
SFRs Modified	None		
Scope:	Driver		
Dependency:	This switch should be set to True only if any of the pins are configured as PortPinDirectionChangeable: TRUE This switch should be set to False if all of the pins are configured as PortPinDirectionChangeable: FALSE		

5.2.4 PortSetPinModeApi

Table 13 PortSetPinModeApi

Name:	PortSetPinModeApi		
Source:	This parameter is defined by the AUTOSAR.		
File:	Port_Cfg.h		
Range:	TRUE - Port_SetPinMode() API is enabled FALSE - Port_SetPinMode() API is disabled		
Default Value:	FALSE		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch STD_ON / STD_OFF used to enable / disable the use of the API Port_SetPinMode (). This configuration parameter is named as PORT_SET_PIN_MODE_API in the configuration file		

SFRs Modified	None
Scope:	Driver
Dependency:	This switch should be set to True only if any of the pins are configured as PortPinModeChangeable: TRUE This switch should be set to False if all of the pins are configured as PortPinModeChangeable: FALSE

5.2.5 PortPBFixedAddress

Table 14 PortPBFixedAddress

Name:	PortPBFixedAddress		
Source:	This parameter is defined by the AUTOSAR.		
File:	Port_Cfg.h		
Range:	TRUE FALSE		
Default Value:	FALSE		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	This configuration parameter allows selecting the Loadable fixed address feature. TRUE – PortPBFixedAddress indicates the configuration parameter set will be loadable fixed address. The feature supports the optimization of a post build configuration with a fixed address by avoiding unnecessary indirection (HisGen0010). The multiplicity of the PortConfigSet is 1..1. FALSE – Loadable fixed address feature is not used. Hence the optimization based on the requirement HisGen0010 is also not supported. The multiplicity of the PortConfigSet is 1..*. This configuration parameter is named as PORT_PB_FIXED_ADDR in the configuration file. The Value of the parameter PORT_PB_FIXED_ADDR is STD_ON / STD_OFF.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5.2.6 PortUserModelInitApiEnable Enable/Disable

Table 15 Enabling/Disabling SFR resetting at Initialization

Syntax:	PortUserModelInitApiEnable
Source:	This is IFX specific parameter
File:	Port_Cfg.h
Range:	Values: TRUE: FALSE: Default Values: FALSE

Configuration class	Pre-compile	X	All variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch to enable / disable protected register access in all the Init API. If PortUserModeInitApiEnable is enabled then the Port Init API will run in User mode. Selection between User Mode 0 and User mode 1 is based on the switch PortRunningInUser0Mode		
SFRs Modified			
Scope:	Driver.		
Dependency:	None		

5.2.7 PortUserModeRuntimeApiEnable Enable/Disable

Table 16 Enabling/Disabling SFR resetting at Initialization

Syntax:	PortUserModeRuntimeApiEnable		
Source:	This is IFX specific parameter		
File:	Port_Cfg.h		
Range:	Values: TRUE: FALSE: Default Values: FALSE		
Configuration class	Pre-compile	X	All variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch to enable / disable protected register access in all the APIs other than Init/Deinit APIs. If PortUserModeRuntimeApiEnable is enabled then all APIs other than Port Init APIs will run in User mode. Selection between User Mode 0 and User mode 1 is based on the switch PortRunningInUser0Mode		
SFRs Modified			
Scope:	Driver.		
Dependency:	None		

5.2.8 PortRunningInUser0Mode Enable/Disable

Table 17 Enabling/Disabling SFR register access in USER0 mode

Syntax:	PortRunningInUser0Mode		
Source:	This is IFX specific parameter		
File:	Port_Cfg.h		
Range:	Values: TRUE: FALSE: Default Values: FALSE		
Configuration	Pre-compile	X	All variants
	Link Time	--	--
	Post build	--	--

class	Link Time	--	--
	Post build	--	--
Description:	If 'PortRunningInUser0Mode' is enabled then the API's selected by 'PortUserModeInitApiEnable' or 'PortUserModeRuntimeApiEnable' will run in user mode 0. If the 'PortRunningInUser0Mode' switch is not selected and if any of the switches 'PortUserModeInitApiEnable', 'PortUserModeRuntimeApiEnable' are selected then those APIs will run in user mode 1. This switch selects between user mode0 and user mode 1 PortRunningInUser0Mode – STD_ON implies User Mode 0 PortRunningInUser0Mode – STD_OFF implies User Mode 1 if any of the switches (PortUserModeInitApiEnable or PortUserModeRuntimeApiEnable) are true else Supervisor Mode (If all the switches mentioned are OFF).		
SFRs Modified			
Scope:	Driver.		
Dependency:	If 'PortRunningInUser0Mode' is enabled then at least any one of the switches 'PortUserModeInitApiEnable' or 'PortUserModeRuntimeApiEnable' should be enabled.		

5.3 Container: PortConfigSet

This container is the base of multiple configuration sets. It contains lists/containers for configuration of Port and Port Pins.

5.3.1 Container: PortContainer

This container is of multiplicity depending on the number of ports in microcontroller. i.e. all the ports are configurable.

5.3.1.1 PortNumberOfPortPins

Table 18 PortNumberOfPortPins

Name:	PortNumberOfPortPins		
Source:	This parameter is defined by the AUTOSAR		
File:	NA		
Range:	1 to 16		
Default value:	Number of pins corresponding to each port.		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	The number of specified PortPins in each PortContainer. It is pre-configured for each PortContainer and can't be changed by the user.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5.3.1.2 PortNumber

Table 19 PortNumber

Name:	PortNumber		
Source:	This parameter is vendor specific		
File:	NA		
Range:	0 to 40 for HE devices 0 to 41 for EP devices		
Default value:	Port number corresponding to each PortContainer.		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	The port number is specified for each PortContainer as a reference to the user. It is pre-configured for each PortContainer and can't be changed by the user.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5.3.1.3 Container: PortPin

Lists down all the parameters required for configuring individual port pins.

The configuration parameters are:

- Pin direction changeable control
- Pin Direction to be initialized
- Pin initial level when set to output
- Pin Initial mode
- Pin mode when direction is changed
- Pin input characteristic
- Pin output characteristic
- Pin symbolic name

The multiplicity of this container is always as the number of pins available in microcontroller and it is expected by the user to configure all used pins. The unused pins will take the default setting.

The PortPin container name can be changed by the user but it is not taken as pin symbolic name. User needs to configure PortPinSymbolicName [5.3.1.3.2](#) parameter for pin symbolic name.

PortPin symbolic name is not generating according to [ecuc_sws_2108] standard. Symbolic name value attribute of Port.bmd configuration entity PortPin is 'false'. Refer [5.3.1.3.2](#) for PortPin Symbolic name generation.

5.3.1.3.1 PortPinId

Table 20 PortPinId

Name:	PortPinId
Source:	This parameter is vendor specific
File:	Port_Cfg.h
Range:	0x000 to 0x28F for HE devices

	0x000 to 0x29B for EP devices.		
Default value:	ID corresponding to the pin		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Unique ID for each port pin. It is pre-configured for each pin and can't be changed by the user. E.g.: The parameter PortPinId = 0x017 (i.e., 23) for Port 1 Pin 7. Here the upper two nibble (bit 4 to bit 11) will represent port number and lower nibble will represent pin number of the port.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5.3.1.3.2 PortPinSymbolicName

Table 21 PortPinSymbolicName

Name:	PortPinSymbolicName		
Source:	This parameter is vendor specific		
File:	Port_Cfg.h		
Range:	ANSI C string		
Default value:	PORT_x_PIN_y where x is port number and y is pin number Ex: For Port0 Pin 0 symbolic value is PORT_0_PIN_0 Port23 Pin 1 symbolic value is PORT_23_PIN_1		
Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Unique name for each port pin. Meaningful name can be given by the user. Other wise pre configured name is taken as the pin symbolic name. Name should start with capital letter. Allowed letters: A-Z, a-z, 0-9, _ PortPinSymbolicName will be #defined with the value of PortPinId in the file Port_Cfg.h. When multiple configurations (PortConfigSet) are used, different PortPinSymbolicName has to be given for the port pins.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5.3.1.3.3 PortPinDirection

Table 22 PortPinDirection

Name:	PortPinDirection		
Source:	This parameter is defined by the AUTOSAR		
File:	Port_PBCfg.c		
Range:	PORT_PIN_IN		

	PORT_PIN_OUT		
Default value:	PORT_PIN_IN		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB
Description:	This parameter allows configuring the direction of each pin. Note: The configuration of this parameter is not valid for analog or digital input ports. The parameter PortPinDirection value is used to derive the parameter PCx (x – pin number) of structure Port_n_locrType. PCx(x – pin number) holds the IOCR register value of the Port pin. PCx = PortPinDirection PortPin(Input/Output)Characteristic PortPinInitialMode If PortPinDirction is PORT_PIN_IN, The bit 4 of the PCx field in the IOCR register is set to 0. If PortPinDirction is PORT_PIN_OUT, The bit 4 of the PCx field in the IOCR register is set to 1.		
SFRs Modified	Px_IOCn(n= 0,4,8,12)		
Scope:	Driver		
Dependency:	None		

5.3.1.3.4 PortPinDirectionChangeable

Table 23 PortPinDirectionChangeable

Name:	PortPinDirectionChangeable		
Source:	This parameter is defined by the AUTOSAR.		
File:	Port_PBCfg.c		
Range:	TRUE - Port Pin direction change allowed FALSE - Port Pin direction change is not allowed		
Default value:	FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB
Description:	Run time parameter allows configuring whether pin direction can be changed during run time or not for each pin. PortPinDirectionChangeable is the member of the structure Port_n_PinType. Notes: - The configuration of this parameter is not valid for analog or digital input ports. - It is advised to configure as direction changeable only for pins whose parameter PortPinMode is configured as PORT_PIN_MODE_GPIO The parameter PortSetPinDirectionApi must be set to True, if this parameter is configured True for any of the port pins		

Name:	PortPinDirectionChangeable
SFRs Modified	None
Scope:	Driver
Dependency:	The parameter PortSetPinDirectionApi must be set to True.

5.3.1.3.5 PortPinInputCharactersticts

Table 24 PortPinInputCharactersticts

Name:	PortPinInputCharactersticts		
Source:	This parameter is microcontroller specific		
File:	Port_PBCfg.c		
Range:	PORT_PIN_IN_NO_PULL PORT_PIN_IN_PULL_DOWN PORT_PIN_IN_PULL_UP		
Default values:	PORT_PIN_IN_PULL_UP		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB
Description:	This parameter allows configuring the input characteristics of each pin. This parameter is used for initialization of the Port module. Note: If PortPinInputCharacteristic is PORT_PIN_IN_NO_PULL, The bit 0 & 1 of the PCx field in the IOCR register is set to 0. If PortPinInputCharacteristic is PORT_PIN_IN_PULL_DOWN, The bit 0 & 1 of the PCx field in the IOCR register is set to 0 & 1 respectively. If PortPinInputCharacteristic is PORT_PIN_IN_PULL_DOWN, The bit 0 & 1 of the PCx field in the IOCR register is set to 1 & 0 respectively.		
SFRs Modified	Px_IOCRn(n= 0,4,8,12)		
Scope:	Driver		
Dependency:	Editing this parameter is allowed only if the AS parameter PortPinDirection is set as PortPinDirectionIn For pins configured for output direction and for read only ports configuration of this parameter is not allowed.		

5.3.1.3.6 PortPinInputHysteresis

Table 25 PortPinInputHysteresis

Name:	PortPinInputHysteresis		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	TRUE – PortPinInputHysteresis enabled FALSE – PortPinInputHysteresis disabled		
Default Value:	FALSE		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD

Description:	This parameter allows configuring Hysteresis support Enable /Disable to portpin. Note: 1. Hysteresis support only for Input pins. 2. Hysteresis support is not available for following portpins. P14.2, P14.4, P14.5, P14.6 and P15.1 Note: Not applicable for TC27XBC device.
Scope:	Module
Dependency:	PortPinDirection = PORT_IN or PortPinDirectionChangable =TRUE

5.3.1.3.7 PortPinOutputCharacteristics

Table 26 PortPinOutputCharacteristics

Name:	PortPinOutputCharacteristics		
Source:	This parameter is microcontroller specific		
File:	Port_PBCfg.c		
Range:	PORT_PIN_OUT_PUSHPULL PORT_PIN_OUT_OPENDRAIN		
Default value:	PORT_PIN_OUT_PUSHPULL		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB
Description:	This parameter allows configuring the output characteristics of each pin. This parameter is used for initialization of the Port module. Note: If PortPinOutputCharacteristic is PORT_PIN_OUT_PUSHPULL, The bit 3 of the PCx field in the IOCR register is set to 0. If PortPinOutputCharacteristic is PORT_PIN_OUT_OPENDRAIN, The bit 3 of the PCx field in the IOCR register is set to 1.		
SFRs Modified	Px_IOCRn(n= 0,4,8,12)		
Scope:	Driver		
Dependency:	Editing this parameter is allowed only if the AS parameter PortPinDirection is set as PortPinDirectionOut For pins configured for input direction and for read only Ports configuration of this parameter is not allowed.		

5.3.1.3.8 PortPinLevelValue

Table 27 PortPinLevelValue

Name:	PortPinLevelValue
Source:	This parameter is defined by the AUTOSAR.
File:	Port_PBCfg.c
Range:	PORT_PIN_LEVEL_HIGH PORT_PIN_LEVEL_LOW

Default value:	PORT_PIN_LEVEL_LOW		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB
Description:	This parameter is to configure the initial pin level for each pin. This level is set to the output latch of the port pin during initialization of the port driver, irrespective of the direction configured to the pin Note: For analog or digital input ports, the configuration of this parameter is not valid.		
SFRs Modified	Px_OMR		
Scope:	Driver		
Dependency:	None		

5.3.1.3.9 PortPinInitialMode

Table 28 PortPinInitialMode

Name:	PortPinInitialMode		
Source:	This parameter is defined by the AUTOSAR		
File:	Port_PBCfg.c		
Range:	PORT_PIN_MODE_GPIO – General purpose input output PORT_PIN_MODE_ALT1 – Alternate output function 1 PORT_PIN_MODE_ALT2 – Alternate output function 2 PORT_PIN_MODE_ALT3 – Alternate output function 3 PORT_PIN_MODE_ALT4 – Alternate output function 4 PORT_PIN_MODE_ALT5 – Alternate output function 5 PORT_PIN_MODE_ALT6 – Alternate output function 6 PORT_PIN_MODE_ALT7 – Alternate output function 7 Note: Supported alternate function names of a respective port pins are given along with ALT1 to ALT7 only for the Tresos configurator “xdm” file. Eg: ALT1 = GTMTOUT0, ALT2 = QSPI0SLSO01, ALT3 = CAN0, ALT4 = ASCLIN10ATX10, ALT5 = PSI5TX1, ALT6 = ERAYTXDA, ALT7 = CCU60CC60		
Default value:	PORT_PIN_MODE_GPIO		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB

Description:	This parameter allows configuring the mode i.e. the different alternate functionality (ALT1, ALT2, ALT3, ALT4, ALT5, ALT6 and ALT7) of each pin during initialization. Port_PinInitialCharacterstics holds the IOCR register value of the Port pin. If the PORT_PIN_MODE_GPIO is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 0, 0 & 0 respectively. If the PORT_PIN_MODE_ALT1 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 0, 0 & 1 respectively. If the PORT_PIN_MODE_ALT2 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 0, 1 & 0 respectively. If the PORT_PIN_MODE_ALT3 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 0, 1 & 1 respectively. If the PORT_PIN_MODE_ALT4 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 1, 0 & 0 respectively. If the PORT_PIN_MODE_ALT5 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 1, 0 & 1 respectively. If the PORT_PIN_MODE_ALT6 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 1, 1 & 0 respectively. If the PORT_PIN_MODE_ALT7 is selected, The bit 0, 1 & 2 of the PCx field in the IOCR register is set to 1, 1 & 1 respectively.
SFRs Modified	Px_IOCRRn(n= 0,4,8,12)
Scope:	Driver
Dependency:	Editing this parameter is not allowed for read only ports

5.3.1.3.10 PortPinModeChangeable

Table 29 PortPinModeChangeable

Name:	PortPinModeChangeable		
Source:	This parameter is defined by the AUTOSAR.		
File:	Port_PBCfg.c		
Range:	TRUE - Port Pin mode change allowed FALSE - Port Pin mode change is not allowed		
Default value:	FALSE		
Configuration Class:	Pre-compile	--	--
	Link Time	--	--
	Post build	x	VariantPB
Description:	Run time parameter allows configuring whether pin mode can be changed during run time or not for each pin. Notes: - The configuration of this parameter is not valid for analog or digital input ports. - The parameter PortSetPinModeApi switch must be set to enabled, for configuring this parameter for any of the port pins.		
SFRs Modified	None		
Scope:	Driver		

Dependency:	The parameter PortSetPinModeApi must be set to True.
-------------	--

5.3.1.3.11 PortPinAnalogInput

Table 30 PortPinAnalogInput

Name:	PortPinAnalogInput		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	ENABLE DISABLE		
Default value:	DISABLE		
Configuration Class	Pre-Compile	--	
	Link Time	--	
	Post Build	x	VariantPB
Description:	This parameter enables configuration of internal Schmitt triggers of the analog inputs as ENABLE or DISABLE. This configuration is used to set Pin Function Decision Control register Px_PDISC. The parameter PortPinAnalogInput value is used to derive the parameter PortFuncDecControlReg of structure Port_n_ConfigType. Each bit of the parameter PortFuncDecControlReg is used as one port pin. If ENABLE is selected the bit value is 1. If DISABLE is selected the bit value is 0. NOTE: Analog input for PORT 0 (P0) pins needs not to be selectable; therefore PortPinAnalogInput is greyed out in Tresos. ADC inputs on P0 can be used without PDISC configuration.		
SFRs Modified	Px_PDISC		
Scope:	Module		
Dependency:	Editable only for Ports which support analog signal feature		

5.3.1.3.12 PortPinDriverStrength

Table 31 PortPinDriverStrength

Name:	PortPinDriverStrength
Source:	This parameter is defined by Infineon Technologies
File:	Port_PBCfg.c
Range:	PORT_CMOS_SPEED_GRADE1 PORT_CMOS_SPEED_GRADE2 PORT_CMOS_SPEED_GRADE3 PORT_CMOS_SPEED_GRADE4 PORT_LVDS_SPEED_GRADE1 [Not applicable for EP devices] PORT_LVDS_SPEED_GRADE2 [Not applicable for EP devices] PORT_LVDS_SPEED_GRADE3 [Not applicable for EP devices] PORT_LVDS_SPEED_GRADE4 [Not applicable for EP devices]
Default value:	PORT_CMOS_SPEED_GRADE4

Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	x	VariantPB
Description:	This parameter allows configuring pin driver strength. Note: - Each Pn_PDR register has PDx and PLx bit fields for x: 0 (8) to 7 (15). Here n corresponds to the port number and x corresponds to the pin number. Two PDR register per port. The parameter PortPinDriverStrength value is used to derive the parameter PadDriverStrengthP0_P7 and PadDriverStrengthP8_P15 of structure Port_n_ConfigType. <u>LVDS / CMOS selection:</u> - Some pins are selectable for channel interface as LVDS or CMOS. - As a CMOS Port where each pin outputs one signal, as any other port (only exception - no open drain mode available) - As an output LVDS port where a pin pair (two pins) outputs one differential MSC signal. - The switching between the two types is done via the PDR register. - For the LVDS type, the pin of the LVDS pair must be programmed as output.		
SFRs Modified	Px_PDR		
Scope:	Module		
Dependency:	None		

5.3.1.3.13 PortPinPadLevel

Table 32 PortPinPadLevel

Name:	PortPinPadLevel		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	PORT_PDR_CMOS_AUTOMOTIVE_LEVEL PORT_PDR_TTL_LEVEL		
Default value:	PORT_PDR_CMOS_AUTOMOTIVE_LEVEL		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	x	VariantPB
Description:	This parameter allows configuring pins pad level. Note: Each Pn_PDR register has PDx and PLx bit fields for x: 0 (8) to 7 (15). Here n corresponds to the port number and x corresponds to the pin number. Two PDR register per port. The parameter PortPinPadLevel value is used to derive the parameter PadDriverStrengthP0_P7 and PadDriverStrengthP8_P15 of structure Port_n_ConfigType.		
SFRs Modified	Px_PDR		

Scope:	Module
Dependency:	None

5.3.1.3.14 PortPinControllerSelect

Table 33 PortPinControllerSelect

Name:	PortPinControllerSelect		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	ENABLE DISABLE		
Default Value:	DISABLE		
Configuration Class	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies Enable / Disable the feature as below, If DISABLE is selected, - It enables Ethernet output via ports alternate output of pin or disable VADC Pull Down Diagnostics/Multiplexer Diagnostics for applicable PortPins. If ENABLE is selected, - It enables Ethernet output via RMII mode or enables VADS VADC Pull Down Diagnostics/Multiplexer Diagnostics for applicable PortPins.		
SFRs Modified	Px_PCSR		
Scope:	Module		
Dependency:	Editable only for Pins supporting this feature.		

5.3.1.3.15 Container: PortPinMode

This container contains the parameter to specify the available modes for the PortPins.

5.4.1.3.14.1 PortPinMode

Table 34 PortPinMode

Name:	PortPinMode
Source:	This parameter is defined by Infineon Technologies
File:	Port_PBCfg.c
Range:	PORT_PIN_MODE_ALL, (Selects all the Modes available for the respective portpin) PORT_PIN_MODE_GPIO, PORT_PIN_MODE_ALT1, PORT_PIN_MODE_ALT2, PORT_PIN_MODE_ALT3, PORT_PIN_MODE_ALT4, PORT_PIN_MODE_ALT5, PORT_PIN_MODE_ALT6, PORT_PIN_MODE_ALT7.
Default Value:	PORT_PIN_MODE_ALL

Configuration Class	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	The parameter PortPinMode will specify the modes that are supported by hardware. This parameter contains a 8-bit value, whose, bit-0 is 1 if PORT_PIN_MODE_GPIO mode is supported, else it is 0 bit-1 is 1 if PORT_PIN_MODE_ALT1 mode is supported, else it is 0 bit-2 is 1 if PORT_PIN_MODE_ALT2 mode is supported, else it is 0 bit-3 is 1 if PORT_PIN_MODE_ALT3 mode is supported, else it is 0 bit-4 is 1 if PORT_PIN_MODE_ALT4 mode is supported, else it is 0 bit-5 is 1 if PORT_PIN_MODE_ALT5 mode is supported, else it is 0 bit-6 is 1 if PORT_PIN_MODE_ALT6 mode is supported, else it is 0 bit-7 is 1 if PORT_PIN_MODE_ALT7 mode is supported, else it is 0 If PORT_PIN_MODE_ALL is selected, the bits are set according to the modes that are available for that portpin. For e.g., if a pin supports GPIO and ALT6 modes, then, bits 0 and 6 are 1 and bit 1,2,3,4,5 and 7 is 0, hence the value of the parameter is 65 (0x41). If he selects PORT_PIN_MODE_ALL for the same pin, then, bits 0 and 6 are 1 and all other bits are zero. i.e., (0x41). Note: This parameter is available in the configuration file in Port_PinSupportedModes data type.		
SFRs Modified	Px_IOCRR(n= 0,4,8,12)		
Scope:	Module		
Dependency:	Editable only if PortSetPinModeApi and PortPinModeChangeable switch is enabled. Its value depends on the Modes Available for the Pin for that derivate..		

5.3.1.4 Container: PortLVDSH

This container specifies the configuration for high performance LVDS.

Multiplicity of PortLVDSH for Port 21 is 2 on target TC27x, TC26x and TC29x

Multiplicity of PortLVDSH for Port 21 is 3 on target TC29x

NOTE: This container is not applicable for EP devices.

5.3.1.4.1 PortLVDSHPinPair

Table 35 PortLVDSHPinPair

Name:	PortLVDSHPinPair		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	NA		
Default Value:	Pin_x_Pin_y, where x and y are the PortPinPair for LVDSH.		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter acts as the parameter to identify the PortPinPair supporting the LVDSH feature.		

	Value is automatically generated from the tool and is non-editable.
SFRs Modified	Px_LPCR
Scope:	Module
Dependency:	Depends on the Port supporting the LVDSH feature.

5.3.1.4.2 PortLVDSRxDisableController

Table 36 PortLVDSRxDisableController

Name:	PortLVDSRxDisableController		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	PORT_CONTROLLED, HSCT_CONTROLLED.		
Default Value:	PORT_CONTROLLED		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies the LVDS enable/disable control function is either PORT controlled or HSCT controlled. If PORT_CONTROLLED is selected, then LVDS enable/disable can be done via Port registers. If HSCT_CONTROLLED is selected then the user should take care of enabling/disabling the LVDS in registers related to HSCT module. This parameter is applicable only for the Rx LVDS pair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins which support LVDS RX feature.		

5.3.1.4.3 PortLVDSRxDisable

Table 37 PortLVDSRxDisable

Name:	PortLVDSRxDisable		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	ENABLE_LVDS, DISABLE_LVDS.		
Default Value:	DISABLE_LVDS		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies whether the PinPair is in LVDSH mode or CMOS mode ENABLE_LVDS disables the CMOS path, DISABLE_LVDS enables the CMOS path. The parameter is applicable only when PortLVDSRxDisableController is PORT_CONTROLLED.		

	This parameter is applicable only for the Rx LVDS pair.
SFRs Modified	Px_LPCR
Scope:	Module
Dependency:	This is editable only for the Pins supporting LVDS RX feature.

5.3.1.4.4 PortLVDSRxTerminationMode

Table 38 PortLVDSRxTerminationMode

Name:	PortLVDSRxTerminationMode		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	EXTERNAL_TERMINATION, INTERNAL_TERMINATION.		
Default Value:	INTERNAL_TERMINATION		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter selects the internal load capacitance, EXTERNAL_TERMINATION, selects external termination on the PCB, INTERNAL_TERMINATION. Selects an 100ohm Receiver internal termination. This parameter is applicable only for the Rx LVDS pair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins supporting LVDS RX feature.		

5.3.1.4.5 PortLV DSTxSignallingMode

Table 39 PortLV DSTxSignallingMode

Name:	PortLV DSTxSignallingMode		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	IEEE_GENERAL_PURPOSE_LINK, IEEE_REDUCED_LVDS_LINK, SPECIAL_REDUCED_LVDS_LINK		
Default Value:	IEEE_GENERAL_PURPOSE_LINK		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter selects the type of electrical characteristics used for Tx LVDS. This parameter is applicable only for the Tx LVDS pair..		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins supporting LVDS TX feature.		

5.3.1.4.6 PortLVDSTxDisableController

Table 40 PortLVDSTxDisableController

Name:	PortLVDSTxDisableController		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	PORT_CONTROLLED, HSCT_CONTROLLED.		
Default Value:	PORT_CONTROLLED		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies the LVDS enable/disable control function is either PORT controlled or HSCT controlled. If PORT_CONTROLLED is selected, then LVDS enable/disable can be done via Port registers. If HSCT_CONTROLLED is selected then the user should take care of enabling/disabling the LVDS in registers related to HSCT module. This parameter is applicable only for the Tx LVDS pair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins supporting LVDS TX feature.		

5.3.1.4.7 PortLVDSTxDisable

Table 41 PortLVDSTxDisable

Name:	PortLVDSTxDisable		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	ENABLE_LVDS, DISABLE_LVDS.		
Default Value:	DISABLE_LVDS		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies whether the PinPair is in LVDSH mode or CMOS mode ENABLE_LVDS: enables LVDS and disables the CMOS path, DISABLE_LVDS: disables LVDS and enables the CMOS path. The parameter is applicable only when PortLVDSTxDisableController is PORT_CONTROLLED. This parameter is applicable only for the Tx LVDS pair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins supporting LVDS TX feature.		

5.3.1.4.8 PortLVDSPowerDown

Table 42 PortLVDSPowerDown

Name:	PortLVDSPowerDown		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	PORT_LVDS_POWER_ON, PORT_LVDS_POWER_DOWN.		
Default Value:	PORT_LVDS_POWER_DOWN		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies the characteristics of the LVDS pair, This parameter is applicable only for the Tx LVDS pair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins supporting LVDS TX feature.		

5.3.1.4.9 PortLVDSPowerDownPullDown

Table 43 PortLVDSPowerDownPullDown

Name:	PortLVDSPowerDownPullDown		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	DISABLE, ENABLE.		
Default Value:	DISABLE		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies the state of the Tx Power down resistor DISABLE, disables the Tx Power down pull down resistor, ENABLE, enables the Tx Power down pull down resistor. This parameter is applicable only for the Tx LVDS pair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	This is editable only for the Pins supporting LVDS TX feature.		

5.3.1.5 Container: PortLVDSM

This container specifies the configuration for medium performance LVDS.
 Multiplicity of PortLVDSM for Port 13 is 2 on target TC27x, TC26x.
 Multiplicity of PortLVDSM for Port 22 is 2 on target TC27x, TC26x, TC29x
 Multiplicity of PortLVDSM for Port 13 is 4 on target TC29x

NOTE: This container is not applicable for EP devices.

5.3.1.5.1 PortLVDSMPinPair

Table 44 PortLVDSMPinPair

Name:	PortLVDSMPinPair		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	NA		
Default Value:	Pin_x_Pin_y, where x and y are the PortPinPair for LVDSM.		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter acts as the parameter to identify the PorPinPair supporting the LVDSM feature Value is automatically generated from the tool and is non-editable.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	Depends on the Port supporting the LVDSM feature.		

5.3.1.5.2 PortPadSupplyVoltage

Table 45 PortPadSupplyVoltage

Name:	PortPadSupplyVoltage		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_PBCfg.c		
Range:	PAD_SUPPLY_3_3V, PAD_SUPPLY_5V		
Default Value:	PAD_SUPPLY_5V		
Configuration class:	Pre-Compile	--	
	Link Time	--	
	Post Build	X	VARIANT-POST-BUILD
Description:	This parameter specifies the level of voltage supply for the LVDSM PortPinPair.		
SFRs Modified	Px_LPCR		
Scope:	Module		
Dependency:	Depends on the Port supporting the LVDSM feature.		

5.4 Container: PortSafety

This container contains the configuration parameters for the safety features of the PORT driver. It is 1:1.

5.4.1 PortSafetyEnable

Table 46 PortSafetyEnable

Syntax:	PortSafetyEnable		
Source:	This parameter is defined by IFX		
File:	Port_Cfg.h		
Range:	Values: TRUE- Safety features enabled for PORT driver FALSE – Safety feature is disabled for PORT driver Default value: FALSE		
Configuration class	Pre-compile	X	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch for enabling the safety features of PORT driver. By enabling this flag, implicitly the following safety features are enabled: • Range check on API parameters is done by PORT driver. • PORT configuration structure will have a unique marker field which will be verified by the driver. The only dependency on marker generation is PortsafetyEnable Configuration parameter. • Safety usage of PORT driver for user to set the respective MPU settings will be published in port_ConfigDoc.html This configuration parameter is named as PORT_SAFETY_ENABLE in the configuration file. The value of the parameter PORT_SAFETY_ENABLE is STD_ON / STD_OFF.		
SFRs Modified	None		
Scope:	Driver.		
Dependency:	None		

5.4.2 PortInitCheckApi

Table 47 PortInitCheckApi

Syntax:	PortInitCheckApi		
Source:	This parameter is defined by IFX		
File:	Port_Cfg.h		
Range:	Values: TRUE - Port_InitCheck() API will be enabled FALSE - Port_InitCheck() API will be disabled Default value: FALSE		
Configuration class	Pre-compile	X	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch for enabling the safety feature Port_InitCheck () API which verifies the initialization done by Port Init API.		

	This configuration parameter is named as PORT_INIT_CHECK_API in the configuration file. The value of the parameter PORT_INIT_CHECK_API is STD_ON / STD_OFF.
SFRs Modified	None
Scope:	Driver
Dependency:	PortInitCheckApi available for edit if PortSafetyEnable switch is STD_ON

5.5 Container: CommonPublishedInformation

This container contains the common published information. These parameters are non-editable.

5.5.1 ArMajorVersion

Table 48 ArMajorVersion

Name:	ArMajorVersion
Source:	This parameter is defined by IFX
File:	Port_Cfg.h
Range:	0 - 255
Default value:	As per the supported Autosar version
Description:	Major version number of AUTOSAR specification on which the appropriate implementation is based on.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.2 ArMinorVersion

Table 49 ArMinorVersion

Name:	ArMinorVersion
Source:	This parameter is defined by IFX
File:	Port_Cfg.h
Range:	0 - 255
Default value:	As per the supported Autosar version
Description:	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.3 ArPatchVersion

Table 50 ArPatchVersion

Name:	ArPatchVersion
Source:	This parameter is defined by IFX

Name:	ArPatchVersion
File:	Port_Cfg.h
Range:	0 - 255
Default value:	As per the supported Autosar version
Description:	Patch version number of AUTOSAR specification on which the appropriate implementation is based on.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.4 SwMajorVersion

Table 51 SwMajorVersion

Name:	SwMajorVersion
Source:	This parameter is defined by IFX
File:	Port_Cfg.h
Range:	0 - 255
Default value:	Refer Port_Cfg.h
Description:	Major version number of the vendor specific implementation of the Driver.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.5 SwMinorVersion

Table 52 SwMinorVersion

Name:	SwMinorVersion
Source:	This parameter is defined by IFX
File:	Port_Cfg.h
Range:	0 - 255
Default value:	Refer Port_Cfg.h
Description:	Minor version number of the vendor specific implementation of the Driver.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.6 SwPatchVersion

Table 53 SwPatchVersion

Name:	SwPatchVersion
Source:	This parameter is defined by IFX

Name:	SwPatchVersion
File:	Port_Cfg.h
Range:	0 - 255
Default value:	Refer Port_Cfg.h
Description:	Patch version number of the vendor specific implementation of the Driver.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.7 ModuleId

Table 54 ModuleId

Name:	ModuleId
Source:	This parameter is defined by IFX
File:	Port.h
Default value:	124
Description:	PORT Driver module ID as described by AUTOSAR.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.8 InstanceId

Table 55 InstanceId

Name:	InstanceId
Source:	This parameter is defined by IFX
File:	Port.h
Default value:	0
Description:	PORT Driver instance ID as described by AUTOSAR.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.9 VendorId

Table 56 VendorId

Name:	VendorId
Source:	This parameter is defined by IFX
File:	Port.h
Default value:	17
Description:	Vendor ID of the dedicated implementation of this Driver according to the AUTOSAR vendor list. Vendor is Infineon Technologies.

Name:	VendorId
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.5.10 VendorApilnfix

Table 57 VendorApilnfix

Name:	VendorApilnfix
Source:	This parameter is defined by IFX
File:	Port.xdm
Default value:	None
Description:	Vendor Specific Name used for extending the API names along with VendorId.
SFRs Modified	None
Scope:	Not Applicable for PORT driver
Dependency:	None

5.5.11 Release

Table 58 Release

Name:	Release
Source:	This parameter is defined by IFX
File:	Port.xdm
Range:	_TRICORE_TC275 _TRICORE_TC277 _TRICORE_TC265 _TRICORE_TC264
Description:	Aurix derivative used for implementation.
SFRs Modified	None
Scope:	Driver
Dependency:	None

5.6 Derived Configuration Parameters

This section describes about the configuration parameters whose values are derived from the values of other parameters configured in the tool. These configuration parameters are available only in the configuration files and not in the tool.

5.6.1 PORTS_AVAILABLE_00_31

Table 59 PORTS_AVAILABLE_00_31

Name:	PORTS_AVAILABLE_00_31
Source:	This parameter is defined by Infineon Technologies

File:	Port_Cfg.h		
Range:	Bit value = 0 implies the port is not available Bit value = 1 implies the port is available Bit 0 is for Port 0, Bit 1 is for Port 1, ... , Bit 31 is for Port 31		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	
	Post Build	--	
Description:	This parameter specifies the ports that are available on the microcontroller		
Scope:	Module		
Dependency:	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.		

5.6.2 PORTS_AVAILABLE_32_63

Table 60 PORTS_AVAILABLE_32_63

Name:	PORTS_AVAILABLE_32_63
Type:	#define
File:	Port_Cfg.h
Description:	This parameter specifies the ports that are available on the microcontroller.
Range	Bit value = 0 implies the port is not available Bit value = 1 implies the port is available Bit 0 is for Port 32, Bit 1 is for Port 33, ... , Bit 9 is for Port 41
Algorithm:	Call Macro CG_GetAvailablePorts32to63
Design Decisions:	None
Asserts/Warnings/Info Raised:	None
Dependencies	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.

5.6.3 PORTS_READONLY_00_31

Table 61 PORTS_READONLY_00_31

Name:	PORTS_READONLY_00_31		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_Cfg.h		
Range:	Bit value = 0 implies the port readable/writable Bit value = 1 implies the port is read only port Bit 0 is for Port 0, Bit 1 is for Port 1, ... , Bit 31 is for Port 31		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	
	Post Build	--	
Description:	Specifies the ports that are read only ports on the microcontroller		
Scope:	Module		

Dependency:	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.
--------------------	---

5.6.4 PORTS_READONLY_32_63

Table 62 PORTS_READONLY_32_63

Name:	PORTS_READONLY_32_63
Type:	#define
File:	Port_Cfg.h
Description:	Specifies the ports that are read only ports on the microcontroller
Range	Bit value = 0 implies the port readable/writable Bit value = 1 implies the port is read only port Bit 0 is for Port32, Bit 1 is for Port 33, ... , Bit 9 is for Port 41
Algorithm:	Call Macro CG_GetReadOnlyPorts32to63
Design Decisions:	None
Asserts/Warnings/Info Raised:	None
Dependencies	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.

5.6.5 PORT_AVAILABLE_PINS_PORT<x>

Table 63 PORT_AVAILABLE_PINS_PORT<x>

Name:	PORT_AVAILABLE_PINS_PORT<x>, x = 0 - 40		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_Cfg.h		
Range:	Bit value = 0 implies the pin is not available Bit value = 1 implies the pin is available Bit 0 is for Pin 0, Bit 1 is for Pin 1, ... , Bit 15 is for Pin 15		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	
	Post Build	--	
Description:	This parameter specifies if the pins of the port are available on the microcontroller		
Scope:	Module		
Dependency:	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.		

5.6.6 PORTS_PCSR_00_31

Table 64 PORTS_PCSR_00_31

Name:	PORTS_PCSR_00_31
Source:	This parameter is defined by Infineon Technologies
File:	Port_Cfg.h
Range:	Bit value = 0 implies the porpin does not supports PCSR feature.

	Bit value = 1 implies the portpin supports PCSR feature Bit 0 is for Port 0, Bit 1 is for Port 1, ... , Bit 31 is for Port 31		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	
	Post Build	--	
Description:	This parameter specifies if the PortPin supports PCSR feature		
Scope:	Module		
Dependency:	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.		

5.6.7 PORTS_PCSR_32_63

Table 65 PORTS_PCSR_32_63

Name:	PORTS_PCSR_32_63
Type:	#define
File:	Port_Cfg.h
Description:	This parameter specifies the ports that supports PCSR feature on the microcontroller from Port32 to Port63.
Range	Bit value = 0 implies the port does not support PCSR feature, Bit value = 1 implies the port supports PCSR feature, Bit 0 is for Port32, Bit 1 is for Port 33, ... , Bit 9 is for Port 41.
Algorithm:	Call Macro CG_GetPCSRPorts32to63
Design Decisions:	None
Asserts/Warnings/Info Raised:	None
Dependencies	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.

5.6.8 PORTS_LVDS_00_31

Table 66 PORTS_LVDS_00_31

Name:	PORTS_LVDS_00_31		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_Cfg.h		
Range:	Bit value = 0 implies the portpin does not supports LVDS feature. Bit value = 1 implies the portpin supports LVDS feature Bit 0 is for Port 0, Bit 1 is for Port 1, ... , Bit 31 is for Port 31		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	
	Post Build	--	
Description:	This parameter specifies if the PortPin supports LVDS feature		
Scope:	Module		
Dependency:	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.		

5.6.9 PORTS_LVDS_32_40

Table 67 PORTS_LVDS_32_40

Name:	PORTS_LVDS_32_40		
Source:	This parameter is defined by Infineon Technologies		
File:	Port_Cfg.h		
Range:	Bit value = 0 implies the portpin does not supports LVDS feature. Bit value = 1 implies the portpin supports LVDS feature Bit 0 is for Port 32, Bit 1 is for Port 33, ... , Bit 8 is for Port 40		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	
	Post Build	--	
Description:	This parameter specifies if the PortPin supports LVDS feature		
Scope:	Module		
Dependency:	The value of this parameter is hardware dependent and depends on the microcontroller derivative used.		

6 Published Parameters

This section describes the published parameters by Port Driver.

6.1 VendorID

Table 68 VendorID

Syntax:	PORT_VENDOR_ID
Type:	Macro
File:	Port.h
Value:	17
Description:	Vendor ID of the dedicated implementation of this Driver according to the AUTOSAR vendor list

6.2 ModuleID

Table 69 ModuleID

Syntax:	PORT_MODULE_ID
Type:	Macro
File:	Port.h
Value:	0x7C
Description:	Module ID of this Driver as described by AUTOSAR

6.3 InstanceID

Table 70 InstanceID

Syntax:	PORT_INSTANCE_ID
----------------	------------------

Type:	Macro - #define, type casted to uint16
File:	Port.h
Value:	0
Description:	This macro gives the Port driver Instance ID as described by AUTOSAR

6.4 ARReleaseMajorVersion

Table 71 ARMajorVersion

Syntax:	PORT_AR_RELEASE_MAJOR_VERSION
Type:	Macro
File:	Port_Cfg.h
Value:	Refer Port_Cfg.h
Description:	Major version number of AUTOSAR specification on which the appropriate implementation is based on.

6.5 ARReleaseMinorVersion

Table 72 ARMinorVersion

Syntax:	PORT_AR_RELEASE_MINOR_VERSION
Type:	Macro
File:	Port_Cfg.h
Value:	Refer Port_Cfg.h
Description:	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

6.6 ARReleaseRevisionVersion

Table 73 ARPatchVersion

Syntax:	PORT_AR_RELEASE_REVISION_VERSION
Type:	Macro
File:	Port_Cfg.h
Value:	Refer Port_Cfg.h
Description:	Patch level version number of AUTOSAR specification on which the appropriate implementation is based on.

6.7 SWMajorVersion

Table 74 SWMajorVersion

Syntax:	PORT_SW_MAJOR_VERSION
Type:	Macro
File:	Port_Cfg.h
Value:	Refer Port_Cfg.h
Description:	Major version number of the vendor specific implementation of the Driver. The numbering is vendor specific.

6.8 SWMinorVersion

Table 75 SWMinorVersion

Syntax:	PORT_SW_MINOR_VERSION
Type:	Macro
File:	Port_Cfg.h
Value:	Refer Port_Cfg.h
Description:	Minor version number of the vendor specific implementation of the Driver. The numbering is vendor specific.

6.9 SWPatchVersion

Table 76 SwPatchVersion

Syntax:	PORT_SW_PATCH_VERSION
Type:	Macro
File:	Port_Cfg.h
Value:	Refer Port_Cfg.h
Description:	Patch level version number of the vendor specific implementation of the Driver. The numbering is vendor specific.

7 API Specification

This chapter covers the type definitions used across the APIs and a description of each API used in the Driver

7.1 API Type Definitions

7.1.1 Type definition Port_ConfigType

Table 77 Type definition Port_ConfigType

Syntax:	Port_ConfigType
Type:	Struct
File:	Port.h
Range:	This is datatype is used for the configuration of the entire Port Driver
Description:	Port configuration structure. The Pointer to the object of this type is passed via Port_Init API to initialize the Port Driver. PortMarker (uint32) – Marker parameter is added for safety.

7.1.2 Type Definition Port_PinType

Table 78 Type Definition Port_PinType

Syntax:	Port_PinType
Type:	uint32
File:	Port.h

Range:	All defined port pins (0x0000 – 0x028F)
Description:	Pin ID is defined with its Port number and the corresponding pin number of Port of type Port_PinType. In LS 16-bit of Pin ID: Bit0 to bit3 represents Pin number Bit4 to bit11 represents Port number Other bits of Pin ID is Reserved For example for Port 5 Pin 15, Pin ID = 0x0000005F

7.1.3 Type Definition Port_PinDirectionType

Table 79 Type Definition Port_PinDirectionType

Syntax:	Port_PinDirectionType
Type:	Enumeration
File:	Port.h
Range:	PORT_PIN_IN PORT_PIN_OUT
Description:	PORT_PIN_IN will set the port pin as input PORT_PIN_OUT will set the port pin as output

7.1.4 Type Definition Port_PinModeType

Table 80 Type Definition Port_PinModeType

Syntax:	Port_PinModeType
Type:	Enumeration
File:	Port.h
Range:	PORT_PIN_MODE_GPIO PORT_PIN_MODE_ALT1 PORT_PIN_MODE_ALT2 PORT_PIN_MODE_ALT3 PORT_PIN_MODE_ALT4 PORT_PIN_MODE_ALT5 PORT_PIN_MODE_ALT6 PORT_PIN_MODE_ALT7
Description:	Pin functionality. PORT_PIN_MODE_GPIO is configurable by Input and Output pin. ALT1 / ALT2 / ALT3 / ALT4 / ALT5 / ALT6 /ALT7 functionality is configurable by only output pins.

7.2 API Functions

[cover parentID=SAS_NAS_ALL_PR455]

This chapter describes the API interfaces provided by Port Driver.

[cover]

7.2.1 Port_Init

Table 81 Specification of API Port_Init

Service Name:	Port_Init
Syntax:	<pre>void Port_Init (const Port_ConfigType *ConfigPtr)</pre>
Service ID:	0x00
Sync / Async:	Synchronous
Origin:	Autosar
Re-entrancy:	Non-reentrant
Parameters (in):	ConfigPtr: Pointer to a configuration
Parameters (out):	None
Return Value:	None
Description:	This initialization function initializes ALL ports and port pins with the configuration set. This function should be called first in order to initialize the port for use. If not called first, then no operation can occur on the MCU ports and port pins. This function will initialize all configured resources. The following rules regarding initialisation of controller registers is apply to the Port Driver implementation: • If the hardware allows for only one usage of the register, the driver implementing that functionality is responsible for initializing the register. • If the register can affect several hardware peripherals and if it is an I/O register it is initialised by this Port Driver. • If the register can affect several hardware peripherals and if it is not an I/O register, it shall be initialised by the MCU Driver. • One-time writable registers that require initialisation directly after reset shall be initialised by the start-up code. The initialization will avoid glitches and spikes on the affected port pins. This function can also be called after a reset, in order to reconfigure the ports and port pins of the MCU. Safety check is enabled if PortSafetyEnable is ON SafeMcal_ReportError service is used for Safety checks.
Caveats:	This service should not be called during a running operation.
Configuration:	All port pins and their functions, and alternate functions can be configured by the configuration tool. It is the user's responsibility to ensure that the same Port/Port pin is not being accessed in parallel by different entities in the same system. Following configuration parameters are used in API PortDevErrorDetect, PortPBFixedAddress, PortSafetyEnable.
DET:	PORT_E_PARAM_CONFIG is reported when passed ConfigPtr parameter is NULL pointer in VariantPB mode.

Service Name:	Port_Init
DEM:	No DEM for Port Driver.
Safety Error reporting	SafeMcal_ReportError will be invoked for following errors: PORT_E_PARAM_CONFIG: When passed configuration pointer parameter is a NULL pointer when PORT_PB_FIXEDADDR is ON or passed pointer is not the desired pointer or Marker value is not matching
Implementation Comments:	Initialization of SFR and global variables. [cover parentID=DS_AS_PORT071] This API is expected to be called after a reset, in order to reconfigure the ports and port pins of the MCU. If not, no operation can occur on the MCU ports and port pins. [cover]
I/O Mode:	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter PortRunningInUser0Mode is 'True' and PortUserModelInitApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModelInitApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModelInitApiEnable is 'False'.

7.2.2 Port_SetPinDirection

Table 82 Specification of API Port_SetPinDirection

Service Name:	Port_SetPinDirection	
Syntax:	void Port_SetPinDirection (Port_PinType Pin, Port_PinDirectionType Direction) 	
Service ID:	0x01	
Sync / Async:	Synchronous	
Origin:	Autosar	
Re-entrancy:	Reentrant for pins of different port or pins assigned to different IOCR register of a same port	
Parameters (in):	Pin	Pin ID
	Direction	Desired direction of the given pin
Parameters (out):	None	
Return Value:	None	
Description:	Sets the port pin direction during runtime. The Port Driver will only allow to changing the direction of those port pins which are configured as changeable.	
Caveats:	This function is re-entrant, if pins of a different port or pins assigned to different IOCR register of a same port are accessed.	
Configuration:	This function is only applicable if the pre-processor switch PORT_PIN_DIRECTION_API is set TRUE. If set to FALSE, this function is not applicable.	

Service Name:	Port_SetPinDirection	
DET:	PORT_E_UNINIT	API is called without initialization.
	PORT_E_PARAM_PIN	Passed Pin parameter is invalid
	PORT_E_DIRECTION_UNCHANGEABLE	Pin is valid but the pin is configured as direction not changeable
	PORT_E_PARAM_DIRECTION (IFX specific for safety)	API Port_SetPinDirection service is called with wrong direction.
DEM:	No DEM for Port Driver.	
Safety Error reporting	SafeMcal_ReportError will be invoked for following errors: • PORT_E_PARAM_PIN: When passed parameter Pin is out of range. • PORT_E_PARAM_DIRECTION: When passed parameter Direction is out of range and is IFX specific for safety.	
Implementation Comments:	Sets port pin direction during runtime whose direction control i.e. PORT_PIN_DIRECTION_CHANGEABLE is configured TRUE. This API is available only if PORT_SET_PIN_DIRECTION_API is switched ON. The port pin direction changeable control parameter i.e. PORT_PIN_DIRECTION_CHANGEABLE is configurable for all pins. The value 0x00 is set if pin direction changeable control is set FALSE. When pin direction change is requested from initialized direction, the pin mode is set as PORT_PIN_MODE_GPIO.	
I/O Mode:	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter PortRunningInUser0Mode is 'True' and PortUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModeRuntimeApiEnable is 'False'.	

7.2.3 Port_RefreshPortDirection

Table 83 Specification of API Port_RefreshPortDirection

Service Name:	Port_RefreshPortDirection
Syntax:	<pre>void Port_RefreshPortDirection (void)</pre>
Service ID:	0x02
Sync / Async:	Synchronous
Origin:	Autosar
Re-entrancy:	Non-reentrant
Parameters (in):	None

Parameters (out):	None
Return Value:	None
Description:	This service refreshes the direction of all configured ports to the configured direction. The Port Driver will exclude those port pins from refreshing that are configured as 'pin direction changeable during runtime'.
Caveats:	None
Configuration:	None
DET:	PORT_E_UNINIT is reported when this API is called before initialization.
DEM:	No DEM for Port Driver.
Implementation Comments:	Input and output pin mode and characteristic as configured for initialization for pins whose direction not changeable will form the refresh value which is generated by the configuration tool.
I/O Mode:	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter PortRunningInUser0Mode is 'True' and PortUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModeRuntimeApiEnable is 'False'.

7.2.4 Port_GetVersionInfo

Table 84 Specification of API Port_GetVersionInfo

Service Name:	Port_GetVersionInfo
Syntax:	void Port_GetVersionInfo (Std_VersionInfoType *VersionInfoPtr)
Service ID:	0x03
Sync / Async:	Synchronous
Origin:	Autosar
Re-entrancy:	Non-reentrant
Parameters (in):	None
Parameters (out):	Versioninfo - Pointer to where to store the version information of this Driver
Return Value:	None
Description:	This service returns the version information of this Driver. The version information includes: - Module Id - Vendor Id - Vendor specific version numbers.
Caveats:	None
Configuration:	This service is only available if enabled by the pre-processor switch PORT_GET_VERSION_INFO_API.

Service Name:	Port_GetVersionInfo	
DET:	PORT_E_PARAM_POINTER	API is called with NULL_PTR parameter.
DEM:	None	
Implementation Comments:	This API is available only if the switch PORT_GET_VERSION_INFO_API is enabled by the user.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter DioRunningInUser0Mode is 'True' and DioUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter DioRunningInUser0Mode is 'False' and DioUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter DioRunningInUser0Mode is 'False' and DioUserModeRuntimeApiEnable is 'False'.	

7.2.5 Port_SetPinMode

Table 85 Specification of API Port_SetPinMode

Service Name:	Port_SetPinMode	
Syntax:	<pre>void Port_SetPinMode (Port_PinType Pin, Port_PinModeType Mode)</pre>	
Service ID:	0x04	
Sync / Async:	Synchronous	
Origin:	Autosar	
Re-entrancy:	Reentrant for pins of different port or pins assigned to different IOCR register of a same port	
Parameters (in):	Pin	Pin ID
	Mode	New port pin mode to be set on port pin
Parameters (out):	None	
Return Value:	None	
Description:	Sets the new port pin mode during runtime.	
Caveats:	This function is re-entrant, if pins of a different port or pins assigned to different IOCR register of a same port are accessed.	
Configuration:	This function is only applicable if the pre-processor switch PortSetPinMode API is set TRUE. If set to FALSE, this function is not applicable.	
DET:	PORT_E_PARAM_PIN	Passed Pin parameter is invalid
	PORT_E_PARAM_INVALID_MODE	Pin is valid but the mode is not valid for the pin.
	PORT_E_UNINIT	API is called without initialization.
	PORT_E_MODE_UNCHANGEABLE	API is called when the mode is unchangeable

DEM:	No DEM for Port Driver.
Safety Error reporting	SafeMcal_ReportError will be invoked for following errors: • PORT_E_PARAM_PIN: When passed parameter Pin is out of range. • PORT_E_PARAM_INVALID_MODE: When passed parameter Mode is out of range.
Implementation Comments:	Sets port pin mode during runtime whose mode control i.e. PORT_PIN_MODE_CHANGEABLE is configured TRUE for that PortPin. Note: Calling this API for input pins will be of no significance since there are no alternate modes for input pin. If called, then the mode will be GPIO as before, but with default input characteristics for the PortPin as, PORT_PIN_IN_NO_PULL. This API is available only if PORT_SET_PIN_MODE_API is switched ON.
I/O Mode:	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter PortRunningInUser0Mode is 'True' and PortUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModeRuntimeApiEnable is 'False'.

7.2.6 Port_InitCheck

Table 86 Specification of API Port_InitCheck

Service Name:	Port_InitCheck	
Syntax:	Std_ReturnType Port_InitCheck (const Port_ConfigType* ConfigPtr)	
Service ID:	None	
Sync / Async:	Synchronous	
Origin:	Vendor specific	
Re-entrancy:	Reentrant	
Parameters (in):	ConfigPtr	Pointer to PORT driver configuration set.
Parameters (out):	None	
Return Value:	E_OK –Initialization is done properly by Port Init API E_NOT_OK – Initialization not properly done by Port Init API	
Description:	This Api checks Port SFR's and Global variables initialized by Port_Init Api.	
Caveats:	Port_Init API can be called in different task before calling InitCheck API	
Configuration:	This API is available only if PortInitCheckApi is TRUE	
DET:	None	
DEM:	No DEM for Port Driver.	

Implementation Comments:	It will check against all SFRs or variables initialized by Port_Init API. It will not modify any SFR/variable, only a read operation is done. It will report an error, if any failure in comparison occurred. Simple if-else condition is used for comparison and returned E_NOT_OK immediately if comparison fails. If failure happens at beginning then returning immediately is best way of saving execution time of InitCheck.
I/O Mode:	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter PortRunningInUser0Mode is 'True' and PortUserModelInitApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModelInitApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter PortRunningInUser0Mode is 'False' and PortUserModelInitApiEnable is 'False'.

7.3 Interrupt Handling

[cover parentID=SAS_NAS_ALL_PR624]

No interrupts are handled by Port Driver.

[/cover]

7.4 Exclusive Area (EA) Information

There is no critical section in Port module.

8 Error Classification

Table 87 Error Classification for DET

Error Code	Error Description	Relevance	Value
PORT_E_PARAM_CONFIG	The Parameter passed is not within range (eg.NULL Pointer)	Development	0x0C
PORT_E_PARAM_PIN	Invalid port pin ID requested	Development	0x0A
PORT_E_DIRECTION_UNCHANGEABLE	Port pin not configured as direction changeable	Development	0x0B
PORT_E_UNINIT	API called before Port_Init	Development	0x0F
PORT_E_PARAM_INVALID_MODE	Invalid port pin Mode requested	Development	0x0D
PORT_E_MODE_UNCHANGEABLE	Port pin not configured as mode changeable	Development	0x0E
PORT_E_PARAM_POINTER	Port_GetVersionInfo called with NULL_PTR	Development	0x10

Table 88 Error Classification for Safety

Type or Error	Error Code	Value
API Port Init service called with wrong parameter	PORT_E_PARAM_CONFIG	0x0C

Type or Error	Error Code	Value
API Port_SetPinMode service called when mode is unchangeable	PORT_E_PARAM_INVALID_MODE	0x0D
Invalid Port Pin ID requested	PORT_E_PARAM_PIN	0x0A
API Port_SetPinDirection service is called with wrong direction.	PORT_E_PARAM_DIRECTION	0x11

9 Example Usage

This chapter describes how the Port Driver can be configured and how to use different API provided by Port Driver.

9.1 Configuration of the Driver

Refer [Table 89](#) for example Pin configuration as Output and direction changeable

Refer [Table 90](#) for example Pin configuration as Input and direction not changeable

For LVDS configuration, the respective PinPair should have PortPinDirection to PORT_PIN_OUT.

Refer [Table 93](#) and [Table 94](#) for Example LVDSH configuration.

Refer [Table 91](#) and [Table 92](#) for Example LVDSM configuration.

Default configuration of protected mode support macros shall be as mentioned below:

PortRunningInUser0Mode -> false

PortUserModeInitApiEnable -> false

PortUserModeRuntimeApiEnable -> false

Table 89 Port 0 Pin 0 Configuration as Output and Direction changeable

Configuration parameter	Value	Description
Pin Container Name	PortPin_0	Pin container name can be changed by the user as same as Pin symbolic name
PortPinId	0	Pin Id is preconfigured and not allowed to change in the configuration tool
PortPinSymbolicName	PortPin_0_GPIO	Port_0_Pin_0 is the preconfigured value but allowed to be changed by user
PortPinDirection	PORT_PIN_OUT	Pin direction configured as Output
PortPinLevelValue	PORT_PIN_LEVEL_HIGH	Initial level of the pin is High after Port Initialization
PortPinInputCharecteristic	PORT_PIN_IN_PULL_DOWN	Configurable since PortPinDirectionChangeable is TRUE. When pin is input, pin characteristic is Pull down
PortPinOutputCharecteristic	PORT_PIN_OUT_PUSH_PULL	When pin is output, pin characteristic is Pushpull
PortPinInitialMode	PORT_PIN_MODE_GPIO	Output mode is general purpose output
PortPinDirectionChangeable	TRUE	Pin direction changeable during runtime using Port_SetPinDirection API
PortPinModeChangeable	FALSE	Pin Mode changeable during runtime

Configuration parameter	Value	Description
		using Port_SetPinMode API
PortPinMode	NA(Since PortPinModeChangeable is FALSE)	Disabled
PortPinPadLevel	PORT_PDR_CMOS_AUTOMOTIVE_LEVEL	Selects the Automotive or CMOS level for Port Pads
PortPinDriverStrength	PORT_CMOS_SPEED_GRADE1	Selects the Speed grade for the Pin

Table 90 Port 0 Pin 1 Configuration as Input and Direction not changeable

Configuration parameter	Value	Description
Pin Container Name	PortPin_1	Pin container name can be changed by the user as same as Pin symbolic name
PortPinId	1	Pin Id is preconfigured and not allowed to change in the configuration tool
PortPinSymbolicName	PortPin_1_GPIO	Port_0_Pin_1 is the preconfigured value but allowed to be changed by user
PortPinDirection	PortPinDirectionIn	Pin direction configured as Input
PortPinLevelValue	NA	Initial level of the pin is not applicable for input pins
PortPinInputCharacteristic	PORT_PIN_IN_PULL_UP	Pin characteristic is Pull up
PortPinOutputCharacteristic	NA	Not applicable for input pin and direction not changeable
PortPinInitialMode	PORT_PIN_MODE_GPIO	Input mode is general purpose input
PortPinDirectionChangeable	FALSE	Pin direction not changeable during runtime
PortPinMode	NA	Disabled

Table 91 LVDSM configuration for Port 13

Configuration Parameter: [Port →PortConfigSet →PortContainer →PortLVDSM]	Value	Description
PortLVDSMPinPair	Pin_0_Pin_1	The Pins which form a LVDSM Pair.
PortPadSupplyVoltage	PAD_SUPPLY_5V	The Voltage on which the transmission over LVDSM should happen.

Table 92 Port 13 Pin 0 configuration for LVDSM

Configuration Parameter: [Port →PortConfigSet →PortContainer →PortPin]	Value	Description
PortPinId	208	Pin Id is preconfigured and not allowed to change in the configuration tool
PortPinSymbolicName	PORT_13_PIN_0	Name populated by the tool by default, can

Configuration Parameter: [Port →PortConfigSet →PortContainer →PortPin]	Value	Description
		be changed by the user. Note: Should be unique across all the config sets.
PortPinDirection	PORT_PIN_OUT	The direction should be OUT for LVDSM.
PortPinDriverStrength*	PORT_LVDS_SPEED_GRADE1	The Speed grade for the Port

Table 93 LVDSH configuration for Port21 PinPair 2 and 3

Configuration Parameter: [Port →PortConfigSet →PortContainer →PortLVDSH]	Value	Description
PortLVDSHPinPair	Pin_2_Pin_3	The Pins which form a LVDSH Pair.
PortLVDSRxDisableController	PORT_CONTROLLED	Port can Enable or Disable the LVDS feature. If HSCT_CONTROLLED is selected then, LVDS feature can be enable or disabled with the SFRs present in HSCT module.
PortLVDSRxDisable	ENABLE_LVDS	Enable LVDS and Disables CMOS.
PortLVDSRXTerminationMode	INTERNAL_TERMINATION	Selects a suitable internal load capacitance

Table 94 LVDSH configuration for Port21 PinPair 4 and 5

Configuration Parameter: [Port →PortConfigSet →PortContainer →PortLVDSH]	Value	Description
PortLVDSHPinPair	Pin_4_Pin_5	The Pins which form a LVDSH Pair.
PortLV DSTxSignallingMode	IEEE_GENERAL_PURPOSE_LINK	
PortLV DSTxDisableController	PORT_CONTROLLED	Port can Enable or Disable the LVDS feature. If HSCT_CONTROLLED is selected then, LVDS feature can be enable or disabled with the SFRs present in HSCT module.
PortLV DSTxDisable	DISABLE_LVDS	Disables LVDS and Enables CMOS.
PortLVDSPowerDown	PORT_LVDS_POWER_DOWN	Selects LVDS Power Down.
PortLVDSPowerDownPullDown	DISABLE	Disables the Power Pull Down feature

The [Code Listing 4](#) provides the example configuration of PORT Driver to initialize the ports

Code Listing 4 Example – configuration for PORT Driver

```

010:  static const Port_n_ConfigType Port_kConfiguration0[] =
011:  {
012:      /* Port 0 */
013:      {
014:          {
015:              /* Port pins direction, characteristics and mode configuration */
016:              {
017:                  ((uint8)PORT_PIN_OUT | PORT_PIN_OUT_PUSH_PULL |
018:                   PORT_PIN_MODE_GPIO), /*Pin 0*/
019:                  ((uint8)PORT_PIN_IN | PORT_PIN_IN_PULL_UP |
020:                   PORT_PIN_MODE_GPIO), /*Pin 1*/
021:                  ... ..
022:                  ... ..
023:                  ((uint8)PORT_PIN_IN | PORT_PIN_IN_PULL_UP |
024:                   PORT_PIN_MODE_GPIO), /*Pin 15*/
025:              }
026:          },
027:          {
028:              /* Port pins initial level configuration */
029:              {
030:                  PORT_PIN_HIGH, /* Pin 0 */
031:                  PORT_PIN_LOW,  /* Pin 1 */
032:                  ... ..
033:                  ... ..
034:                  PORT_PIN_LOW,  /* Pin 15 */
035:              }
036:          },
037:          /* Port pins drive strength configuration */
038:          Port0_lPdrConfig(
039:              (PORT_PDR_CMOS_AUTOMOTIVE_LEVEL |
040:               PORT_CMOS_SPEED_GRADE1), /* Pin 0 */
041:              ... ..
042:              ... ..
043:          ),
044:          {
045:              /* Port pin run time direction changeable or not configuration */
046:              {
047:                  PORT_PIN_DIR_CHANGEABLE, /* Pin 0 */
048:                  PORT_PIN_DIR_NOT_CHANGEABLE, /* Pin 1 */
049:                  ... ..
050:                  PORT_PIN_DIR_NOT_CHANGEABLE, /* Pin 15 */
051:              }
052:          },
053:          {
054:              /* Port pins direction, characteristics and mode configuration if
055:               direction is changeable */
056:              {
057:                  ((uint8)PORT_PIN_IN | PORT_PIN_IN_PULL_DOWN |
058:                   PORT_PIN_MODE_GPIO), /*Pin 0*/

```

Code Listing 4 Example – configuration for PORT Driver

```

059:          ((uint8)PORT_PIN_IN | PORT_PIN_IN_PULL_UP |
060:            PORT_PIN_MODE_GPIO), /*Pin 1*/
061:          ... ..
062:          ... ..
063:          ((uint8)PORT_PIN_IN | PORT_PIN_IN_PULL_UP |
064:            PORT_PIN_MODE_GPIO), /*Pin 1*/
065:      }
066:  }
067: },
068: /* Port 1 */
069: ... ..
070: ... ..
071: } ;
072:
073: static const Port_n_LVDSConfigType Port_kLVDSConfig0[] =
074: {
075:     { /*          Port13          */
076:         Port_lLvdsM(
077:             PORT_LVDS_PAD_SUPPLY_5_VOLTAGE
078:         ),
079:         ... ..
080:     },
081:     { /*          Port21          */
082:         PORT_LVDSH_DEFAULT,
083:         Port_lLvdsH(
084:             PORT_LVDS_PORT_CONTROLLED,
085:             PORT_LVDS_ENABLE,
086:             PORT_LVDS_INTERNAML_TERMINATION,
087:             PORT_LVDS_POLY_RESISTOR_TERM,
088:             PORT_LVDS_DEFAULT,
089:             PORT_LVDS_DEFAULT,
090:             PORT_LVDS_DEFAULT,
091:             PORT_LVDS_DEFAULT,
092:             PORT_LVDS_DEFAULT
093:         ),
094:         Port_lLvdsH(
095:             PORT_LVDS_DEFAULT,
096:             PORT_LVDS_DEFAULT,
097:             PORT_LVDS_DEFAULT,
098:             PORT_LVDS_DEFAULT,
099:             PORT_LVDS_IEEE_GEN_PURPOSE_LINK,
100:             PORT_LVDS_PORT_CONTROLLED,
101:             PORT_LVDS_DISABLE,
102:             PORT_LVDS_POWER_DOWN,
103:             PORT_LVDS_POWDOWN_PULDOWN_DISABLE
104:         )
105:     },
106:     ... ..
107: },

```


Code Listing 4 Example – configuration for PORT Driver

```

0108:  }
0109:
0110:
0111:  const Port_ConfigType Port_ConfigRoot [1] =
0112:  {
0113:  {
0114:      /* Port Marker used for checking configuration */
0115:      #if (PORT_SAFETY_ENABLE == STD_ON)
0116:      const uint32 PortMarker;
0117:      #endif
0118:      ... ..
0119:      /* Port Set Configuration */
0120:      & Port_kConfiguration0[0],
0121:      /* LVDS configuration */
0122:      &Port_kLVDSConfig0[0],
0123:      ... ..
0124:  }
0125:  };

```

9.2 Initialization of Port Driver

Initialization of Port Driver is done by calling API Port_Init.

An example below shows call to Port_Init API which will initialize the Port configuration.

Code Listing 5 Initialization of Port Driver

```

0126:  Port_Init(&Port_ConfigRoot[0]);

```

Port_Init can be called without any de-initialization. This function may also be used to initialize the Driver software and reinitialize the ports and port pins to another configured state depending on the configuration set passed to this service.

Note:

User must take care that the Port_Init API is called before using any other API provided by Port Driver. Development error is reported if user calls other Port Driver's API before calling Port_Init.

9.3 Using the API

9.3.1 Changing the Direction of the Pin

The Port Driver provides the API to change the direction of the pin during run time. User must enable 'PortPinDirectionChangeable' for the required pin. API Port_SetPinDirection is used to change the direction.

This API Port_SetPinDirection is available only when the 'PortSetPinDirectionApi' is switched ON. If API is called with 'PortSetPinDirectionApi' is switched OFF, then there will compilation error "ERROR_Port_SetPinDirection_NOT_SELECTED".

Code Listing 6 Changing the Direction of Pin

```

001 :Port_SetPinDirection (PortPin_0_GPIO, PORT_PIN_IN);

```

Symbolic name PortPin_0_GPIO represents Pin 0 of Port 0. While changing the direction, the other characteristics of the pin are taken from the configuration mentioned in [Table 89](#). By calling to API as mentioned in [Code Listing 6](#) Pin 0 of Port 0 will be changed to Input direction with pull down characteristics.

9.3.2 Refreshing Direction of all Configured Pins

The Port Driver provides the API to refresh the direction of all the configured pins. This API does not refresh the direction of pins, where pin direction is configured as changeable

Code Listing 7 Refreshing Direction of all Configured Pins

```
001 :Port_RefreshPortDirection();
```

9.3.3 Set new pin mode

The Port Driver provides the API to runtime change the pin alternate functionality. This API does not change the pin direction.

This API is available only when the 'PortSetPinModeApi' is switched ON. If API is called with 'PortSetPinModeApi' is switched OFF, then there will be a compilation error "ERROR_Port_SetPinMode_NOT_SELECTED".

Code Listing 8 Switching pin alternate mode

```
001 : Port_SetPinMode(PORT_0_PIN_0, PORT_PIN_MODE_ALT1);
```

9.3.4 Getting Version Information

The Port Driver provides the API to store the version information of this Driver at the specified location. The version information includes:

- Module ID
- Vendor ID
- Vendor specific version numbers

This function Port_GetVersionInfo is available with the pre-compiler switch PortVersionInfoApi set to ON. If API is called with PortVersionInfoApi is switched OFF, then there will compilation error "ERROR_Port_GetVersionInfo_NOT_SELECTED".

Code Listing 9 Getting Version Information of the Driver

```
001 :Port_GetVersionInfo(&VersionInfo_Location);
```

VersionInfo_Location is of type Std_VersionInfoType. It is the place for the API to store the version information

10 Assumptions and Limitations

10.1 Assumptions and Deviations from Software Specification

- The requirement PORT003: The function Port_Init() may also be used to initialize the driver software and reinitialize the ports and port pins to another configured state depending on the configuration set passed to this service.
- For Direction changeable Pins whose parameter PortPinInitialMode is configured as GPIO, the mode remains as GPIO when the pin is switched between directions input and output using Port_SetPinDirection API. For Pins whose parameter PortPinInitialMode is configured as any of the alternate modes, the alternate mode is valid only when the direction is output. When the direction is switched to input the mode automatically is made as GPIO.
- If the PortDevErrorDetect is OFF and PortPinDirectionChangeable switch is OFF, then the Port Pin direction will not be changed upon calling Port_SetPinDirection () API. Also, if the PortDevErrorDetect is OFF and PortPinModeChangeable switch is OFF, then the Port Pin Mode will not be changed upon calling Port_SetPinMode () API. Refer Bugzilla ID #12065.

- PortPin symbolic name is not generating according to [ecuc_sws_2108] standard. Symbolic name value attribute of Port.bmd configuration entity PortPin is 'false'.Refer [5.3.1.3.2](#) for PortPin Symbolic name generation.

10.2 Limitations and Deviations

- The Feature Called LVDS is not applicable for EP (23x) devices.
- Hysteresis Support is not available for EP devices and TC27XBC device.
- Deviation from Autosar. Limited re-entrancy is at IOCR register level. i.e., Port pins having different IOCR registers can be accessed concurrently. Port_SetPinDirection() or Port_SetPinMode() API shall not be called concurrently for the pins shares the same IOCR register.

10.3 Hardware features NOT Supported

The table below lists down all the hardware features that are not supported by the Port Driver. These features are out of the scope of Port Driver's specification.

Note:

- P20.2 is a read only pin, the configuration gives both output and input for this pin, however if configured for output an error is thrown.
- LVDSH and LVDSM feature is not supported by EP (23X) devices.
- Hysteresis feature is not supported by EP (23X) devices and TC27XBCdevice.

Table 95 Hardware Features not supported

Emergency Stop functionality provided by GPIO hardware unit for all the pins.

11 Used Aurix Deriatives

Table 96 Aurix Derivatives Information

Derivative	Information	Feature	Availability
TC2Dx	Number of Ports: 18 Number of I/O Pins: 177 Number of input only pins: 10	Port 00	Available pins: 0 – 12
		Port 01	Available pins: 0 – 13
		Port 02	Available pins: 0 – 15
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0 – 8
		Port 11	Available pins: 0 – 15
		Port 12	Available pins: 0 – 1
		Port 13	Available pins: 0 – 3
		Port 14	Available pins: 0 – 10
		Port 15	Available pins: 0 – 8
		Port 16	Not available

		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0 – 3, 6 - 14
		Port 21	Available pins: 0 – 7
		Port 22	Available pins: 0 – 11
		Port 23	Available pins: 0 – 7
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Available pins: 0 – 7
		Port 31	Not available
		Port 32	Available pins: 0 – 6
		Port 33	Available pins: 0 – 15
		Port 34	Available pins: 0 – 10
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0 – 9
TC277	Number of Ports: 17 Number of I/O Pins: 160 Number of input only pins: 10	Port 00	Available pins: 0 – 12
		Port 01	Available pins: 3 – 7
		Port 02	Available pins: 0 – 11
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0 – 8
		Port 11	Available pins: 0 – 15
		Port 12	Available pins: 0 – 1
		Port 13	Available pins: 0 – 3
		Port 14	Available pins: 0 – 10
		Port 15	Available pins: 0 – 8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0 – 3, 6 – 14

		Port 21	Available pins: 0 – 7
		Port 22	Available pins: 0 – 11
		Port 23	Available pins: 0 – 7
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Available pins: 0, 2 – 7
		Port 33	Available pins: 0 – 15
		Port 34	Available pins: 1 – 5
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0 – 9
TC275	Number of Ports: 14 Number of I/O Pins: 121 Number of input only pins: 10	Port 00	Available pins: 0 – 12
		Port 01	Not available
		Port 02	Available pins: 0 – 8
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0 – 8
		Port 11	Available pins: 2-3, 6, 9-12
		Port 12	Not available
		Port 13	Available pins: 0 – 3
		Port 14	Available pins: 0 – 10
		Port 15	Available pins: 0 – 8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0 – 3, 6 – 14
		Port 21	Available pins: 0 – 7
		Port 22	Available pins: 0 – 3
		Port 23	Available pins: 0 – 5
		Port 24	Not available

		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Available pins: 0, 2 – 4
		Port 33	Available pins: 0 – 13
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0 – 9
TC264	Number of Ports: 14 Number of I/O Pins: 93 Number of input only pins: 6	Port 00	Available pins: 0-9,12
		Port 01	Not available
		Port 02	Available pins: 0-8
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 1-3,5-6
		Port 11	Available pins: 2-3,6,9-12
		Port 12	Not available
		Port 13	Available pins:0-3
		Port 14	Available pins: 0-6
		Port 15	Available pins: 0-8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0,2-3,6-14
		Port 21	Available pins: 2-7
		Port 22	Available pins: 0-3
		Port 23	Available pins: 1
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available

		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Available pins: 0,4
		Port 33	Available pins: 4-13
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-1,6-9
TC265	Number of Ports: 14 Number of I/O Pins: 119 Number of input only pins: 10	Port 00	Available pins: 0-12
		Port 01	Not available
		Port 02	Available pins: 0-8
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0-8
		Port 11	Available pins: 2-3,6,9-12
		Port 12	Not available
		Port 13	Available pins:0-3
		Port 14	Available pins: 0-10
		Port 15	Available pins: 0-8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0-3,6-14
		Port 21	Available pins: 0-7
		Port 22	Available pins: 0-3
		Port 23	Available pins: 0-5
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Available pins: 0,2-4

		Port 33	Available pins: 0-13
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-9
TC299	Number of Ports: 22 Number of I/O Pins: 276 Number of input only pins: 15	Port 00	Available pins: 0-15
		Port 01	Available pins: 0-15
		Port 02	Available pins: 0-15
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0-11,13-15
		Port 11	Available pins: 0-15
		Port 12	Available pins: 0-1
		Port 13	Available pins: 0-7,9-15
		Port 14	Available pins: 0-15
		Port 15	Available pins: 0-8,10-15
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0-3,6-14
		Port 21	Available pins: 0-7
		Port 22	Available pins: 0-11
		Port 23	Available pins: 0-7
		Port 24	Available pins: 0-15
		Port 25	Available pins: 0-15
		Port 26	Available pins: 0
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Available pins: 0-15
		Port 31	Available pins: 0-15
		Port 32	Available pins: 0,2-7
		Port 33	Available pins: 0-15
		Port 34	Available pins: 1-5
		Port 35	Not available
		Port 36	Not available

TC298	Number of Ports: 22 Number of I/O Pins: 243 Number of input only pins: 12	Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-14
		Port 00	Available pins: 0-15
		Port 01	Available pins: 0-15
		Port 02	Available pins: 0-15
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0-11,13-15
		Port 11	Available pins: 0-15
		Port 12	Available pins: 0-1
		Port 13	Available pins: 0-7,9-14
		Port 14	Available pins: 0-15
		Port 15	Available pins: 1 - 8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0,2
		Port 21	Available pins: 0-7
		Port 22	Available pins: 0-3
		Port 23	Available pins: 0-6
		Port 24	Available pins: 0-15
		Port 25	Available pins: 0-5,7-15
		Port 26	Available pins: 0
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Available pins: 0-15
		Port 31	Available pins: 0-15
		Port 32	Available pins: 0,2-7
		Port 33	Available pins: 0-15
		Port 34	Available pins: 1-5
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-8,11-14

TC233/TC223	Number of Ports: 13 Number of I/O Pins: 78	Port 00	Available pins: 0
		Port 01	Not available
		Port 02	Available pins: 0-8
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 5-6
		Port 11	Available pins: 2,3,6,8-12
		Port 12	Not available
		Port 13	Available pins: 0-3
		Port 14	Available pins: 0,1,3,4,6
		Port 15	Available pins: 0,1,2,3,5
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 2,8,9,10,11,12,13,14
		Port 21	Available pins: 2,3,4,6,7
		Port 22	Not available
		Port 23	Available pins: 1
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Not available
		Port 33	Available pins: 5-10
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-11
		Port 41	Available pins: 0-11

TC234/TC237/TC224	Number of Ports: 15 Number of I/O Pins: 120	Port 00	Available pins: 0-9,12
		Port 01	Not available
		Port 02	Available pins: 0-8
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 1-3,5-6
		Port 11	Available pins: 2,3,6,8-12
		Port 12	Not available
		Port 13	Available pins: 0-3
		Port 14	Available pins: 0-8
		Port 15	Available pins: 0-8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0,2,3,6-14
		Port 21	Available pins: 2-7
		Port 22	Not available: 0-4
		Port 23	Available pins: 1
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Not available
		Port 33	Available pins: 0-12
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-11
		Port 41	Available pins: 0-11

TC267	Number of Ports: 14 Number of I/O Pins: 119 Number of input only pins: 10	Port 00	Available pins: 0-12
		Port 01	Not available
		Port 02	Available pins: 0-8
		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 0-8
		Port 11	Available pins: 2-3,6,9-12
		Port 12	Not available
		Port 13	Available pins:0-3
		Port 14	Available pins: 0-10
		Port 15	Available pins: 0-8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0-3,6-14
		Port 21	Available pins: 0-7
		Port 22	Available pins: 0-3
		Port 23	Available pins: 0-5
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Available pins: 0,2-4
		Port 33	Available pins: 0-13
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-9
TC222	Number of Ports: 12 Number of I/O Pins: 59	Port 00	Available pins: 0
		Port 01	Not available
		Port 02	Available pins: 0-8

		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 5-6
		Port 11	Available pins: 2,3,6,9-12
		Port 12	Not available
		Port 13	Not available
		Port 14	Available pins: 0,1,3,4,6
		Port 15	Available pins: 0,1,2,3,5
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 2,8,9,11,12,13,14
		Port 21	Available pins: 6,7
		Port 22	Not available
		Port 23	Available pins: 1
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Not available
		Port 33	Available pins: 5-10
		Port 34	Not available
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 5-11
		Port 41	Available pins: 0-6

TC297	Number of Ports: 17 Number of I/O Pins: 159	Port 00	Available pins: 0-12
		Port 01	Available pins: 3-7
		Port 02	Available pins: 0-11

		Port 03	Not available
		Port 04	Not available
		Port 05	Not available
		Port 06	Not available
		Port 07	Not available
		Port 08	Not available
		Port 09	Not available
		Port 10	Available pins: 1-8
		Port 11	Available pins: 0-15
		Port 12	Available pins: 0-1
		Port 13	Available pins: 0-3
		Port 14	Available pins: 0-10
		Port 15	Available pins: 0-8
		Port 16	Not available
		Port 17	Not available
		Port 18	Not available
		Port 19	Not available
		Port 20	Available pins: 0-3,6-14
		Port 21	Available pins: 0-7
		Port 22	Available pins: 0-11
		Port 23	Available pins: 0-7
		Port 24	Not available
		Port 25	Not available
		Port 26	Not available
		Port 27	Not available
		Port 28	Not available
		Port 29	Not available
		Port 30	Not available
		Port 31	Not available
		Port 32	Available pins: 0, 2-7
		Port 33	Available pins: 0-15
		Port 34	Available pins: 1-5
		Port 35	Not available
		Port 36	Not available
		Port 37	Not available
		Port 38	Not available
		Port 39	Not available
		Port 40	Available pins: 0-9
		Port 41	Not available

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